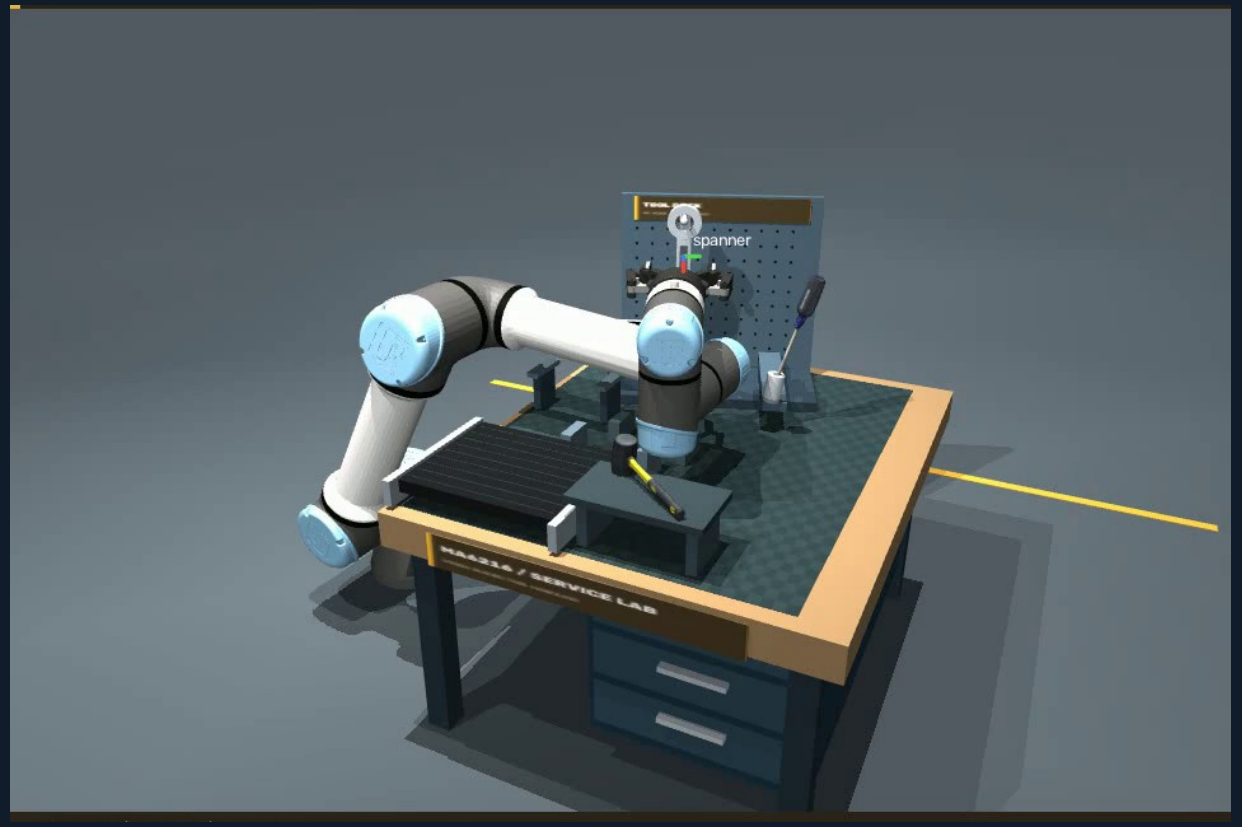
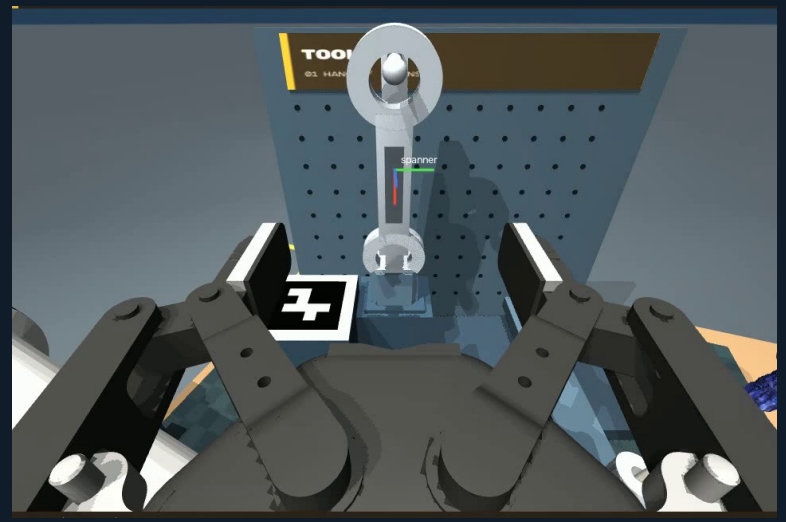
From visual observations to physical tool handling



OVERVIEW / VIDEO 195.00 s



WRIST / VIDEO 195.00 s

4 tools

6D object poses

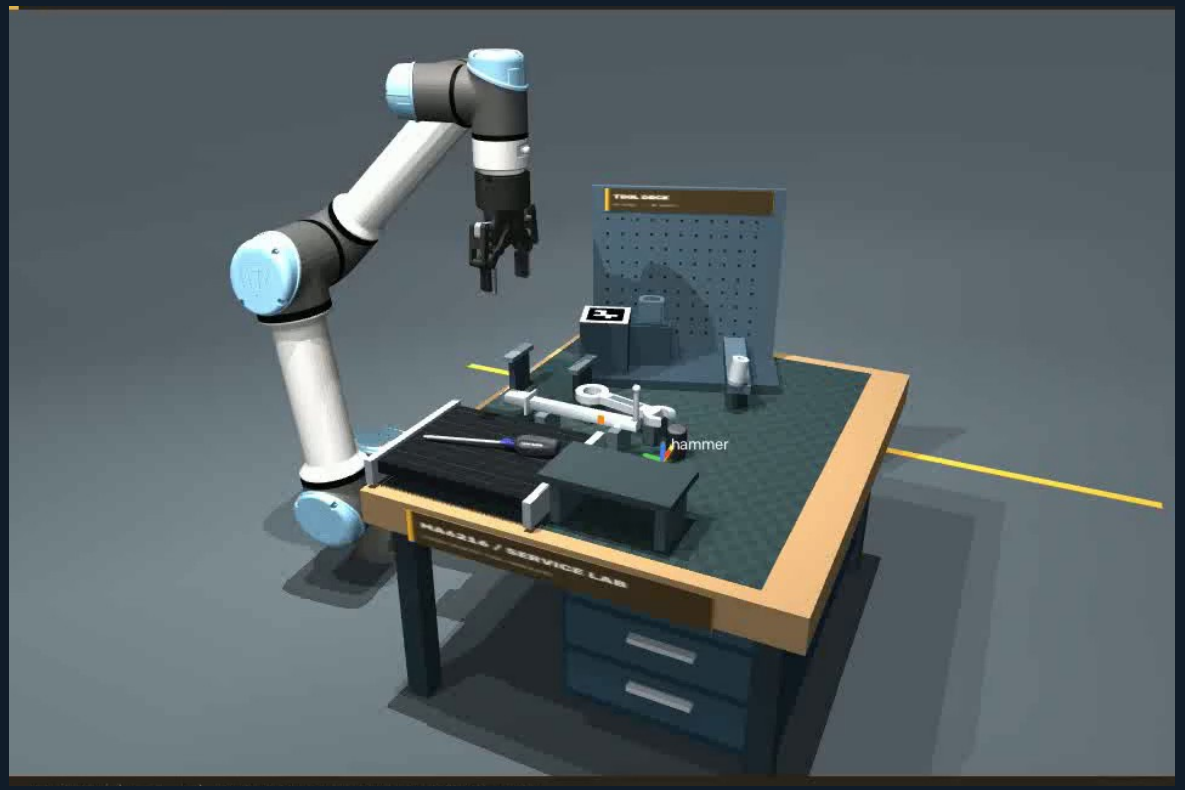
Physical release + support

Vision-Guided Robotic Manipulation

APPLICATION

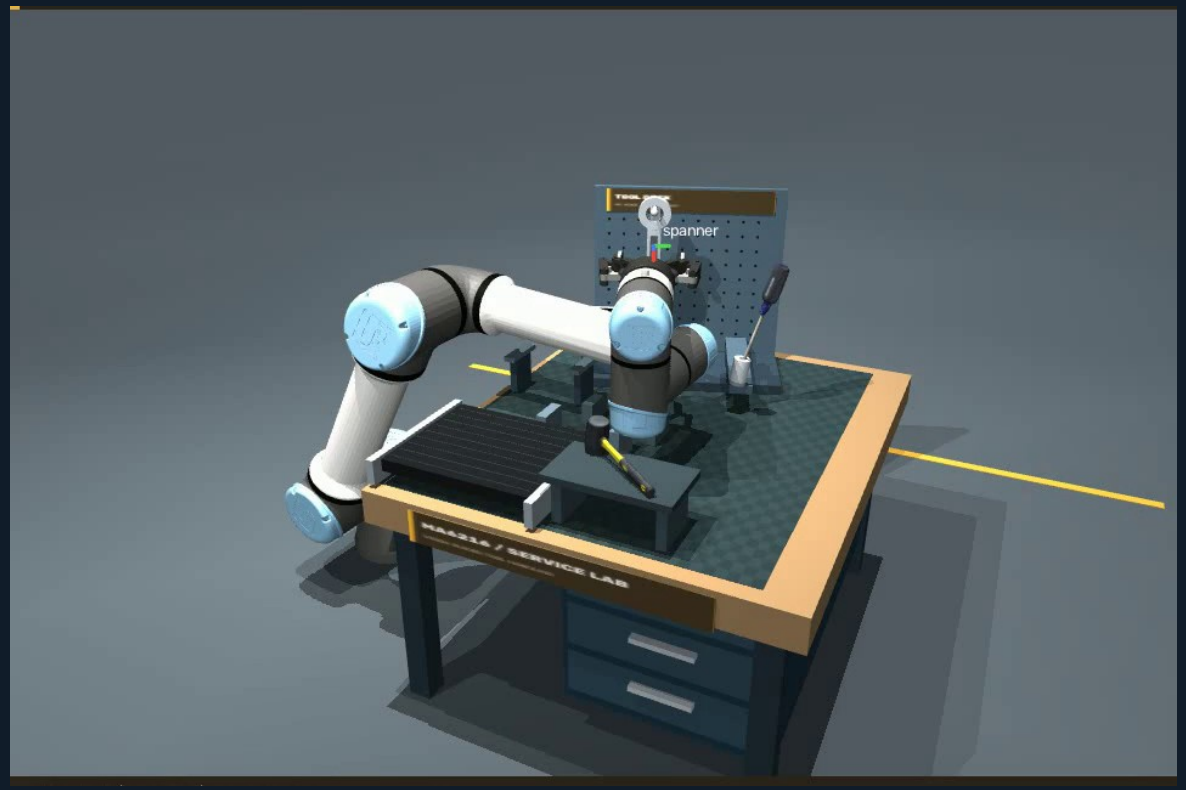
One continuous scene: before and after

START / tools at their pickup stations



OVERVIEW / VIDEO 0.00 s

FINISH / all four final checks pass



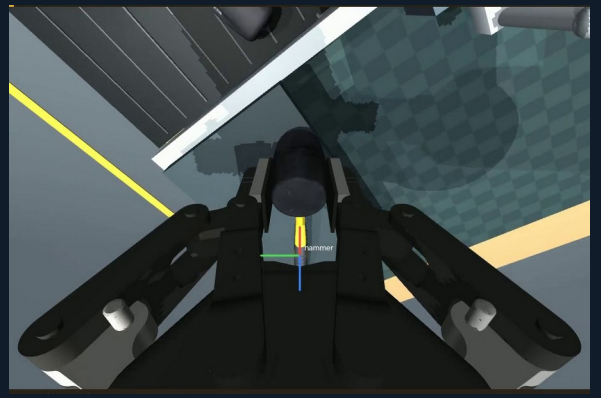
OVERVIEW / VIDEO 195.00 s

The task ends after release and stability checks.

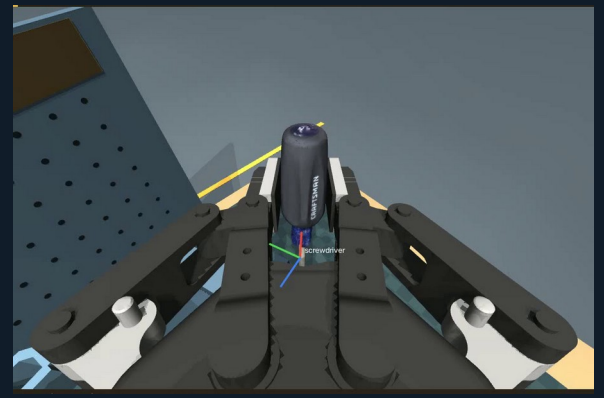
Final 48 trials | iterative / no disturbance / seed 400 | playback timestamps

TASK GEOMETRY

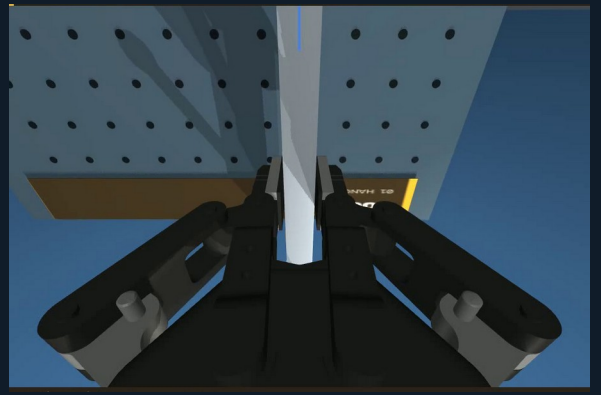
Four operations that depend on orientation



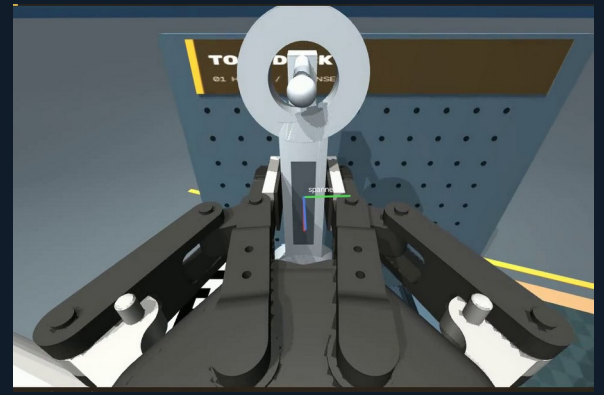
01 Return the hammer



02 Insert the tilted screwdriver



03 Install the independent rack



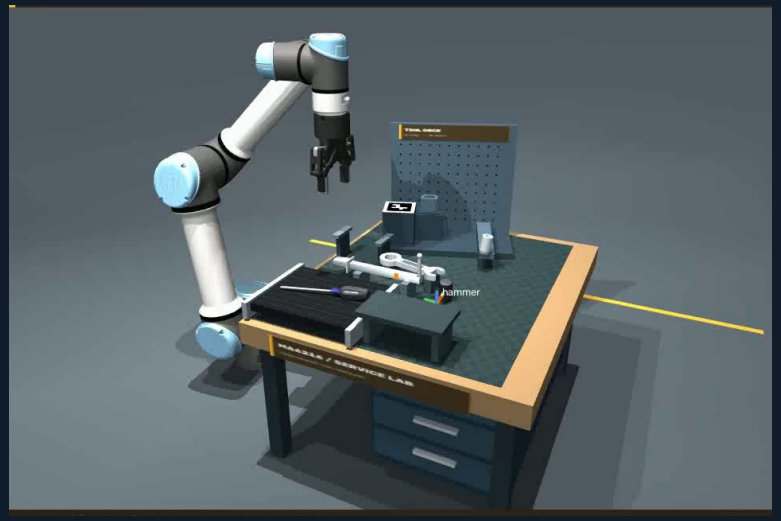
04 Regrasp and thread the spanner

Position locates the tool. Orientation makes the operation possible.

SYSTEM ARCHITECTURE

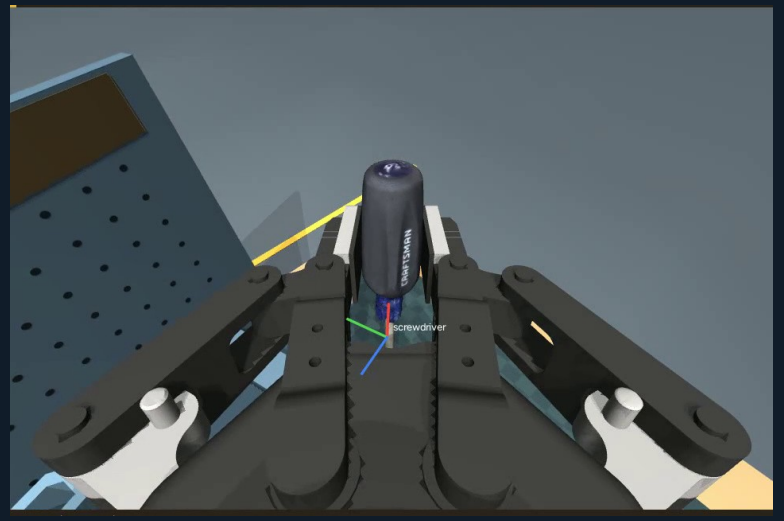
An image-to-action chain you can inspect

RGB-D observation



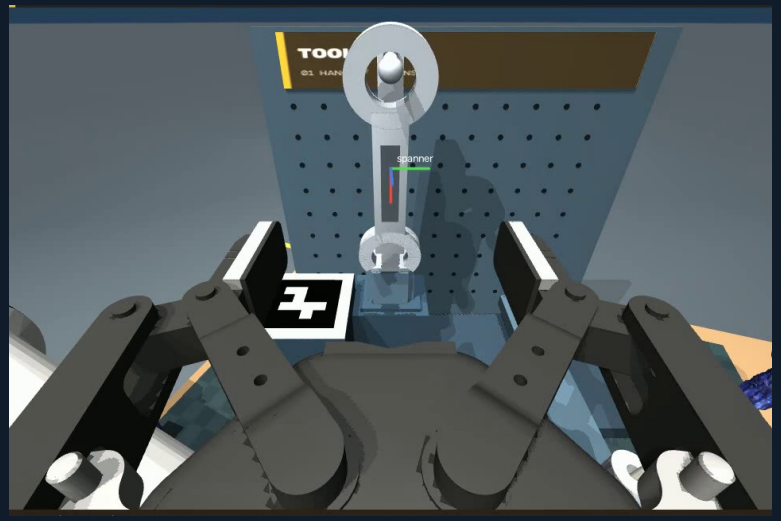
OVERVIEW / VIDEO 0.00 s

6D pose in the wrist view



WRIST / VIDEO 54.40 s

Released physical outcome



WRIST / VIDEO 195.00 s

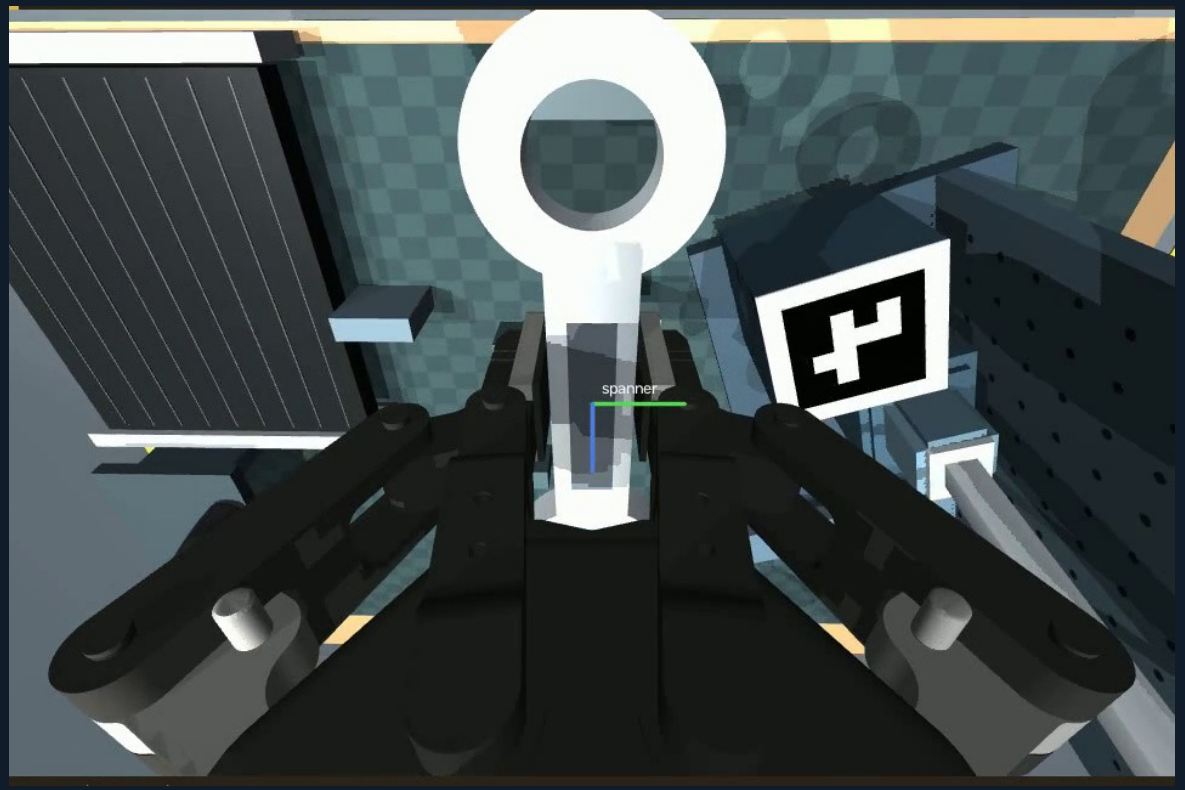
RGB-D → mask → camera_T_object → robot target → control

Reobserve → bounded correction | Simulation truth → evaluation only

Image estimates guide the robot; independent checks score the result.

TASK 1 / SELECTION

Technical choices follow the manipulation task



WRIST / VIDEO 158.20 s

FoundationPose

6D pose from reconstructed RGB-D references

MegaPose / GraspNet

Reviewed pose and grasp alternatives; not external experimental baselines

MuJoCo + OpenCV

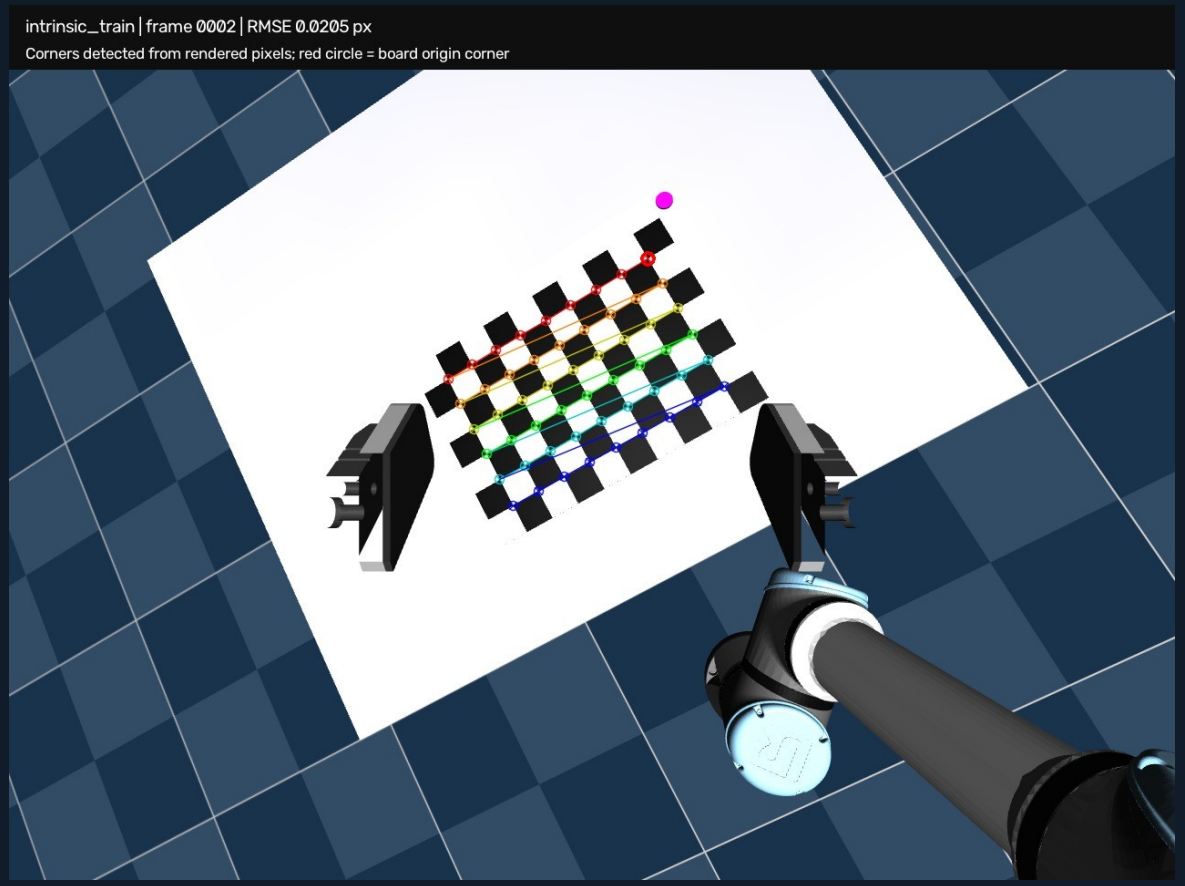
Reuse contact dynamics; estimate cameras from images

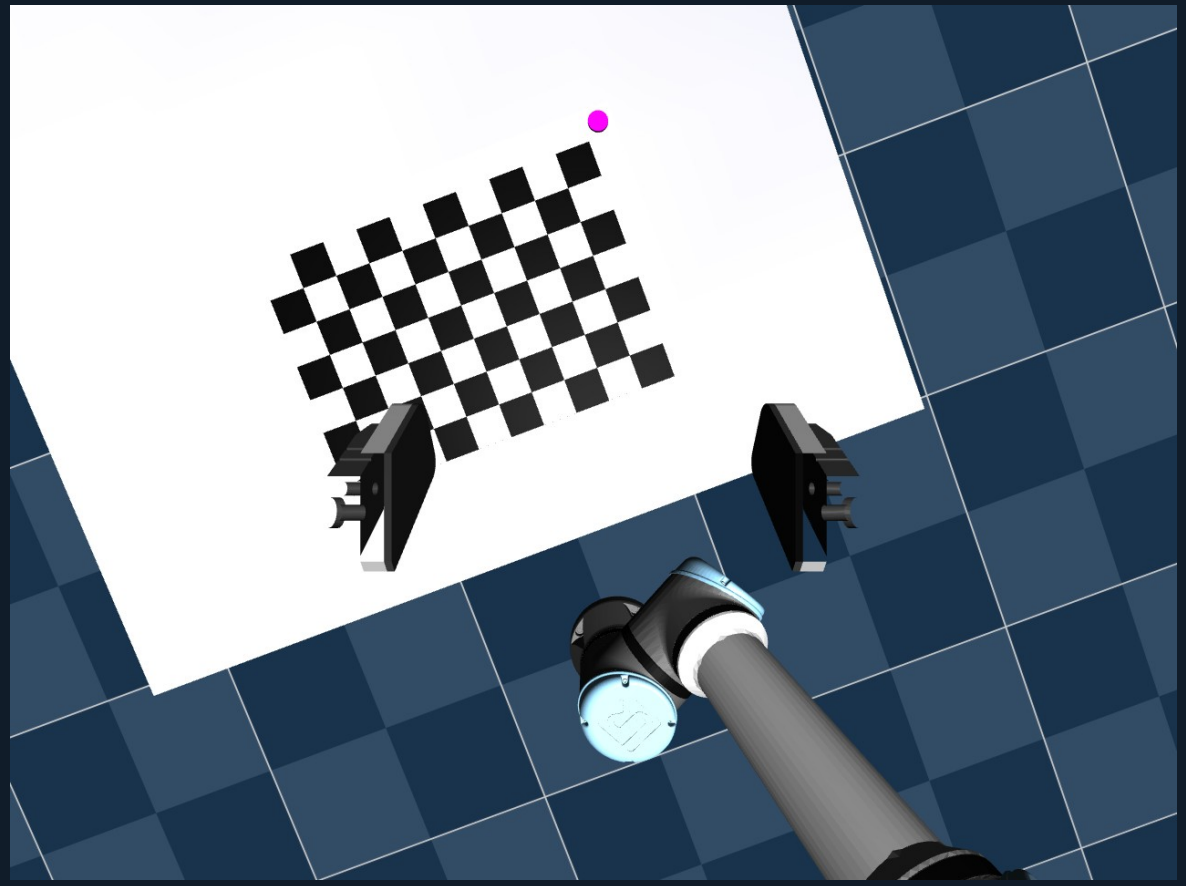
Model-free here does not mean reference-free.

Wen et al. 2024; Labbé et al. 2022; Fang et al. 2020; Todorov et al. 2012 | literature review in repository

TASK 2 / CAMERA INTRINSICS

Calibrate from rendered checkerboard pixels

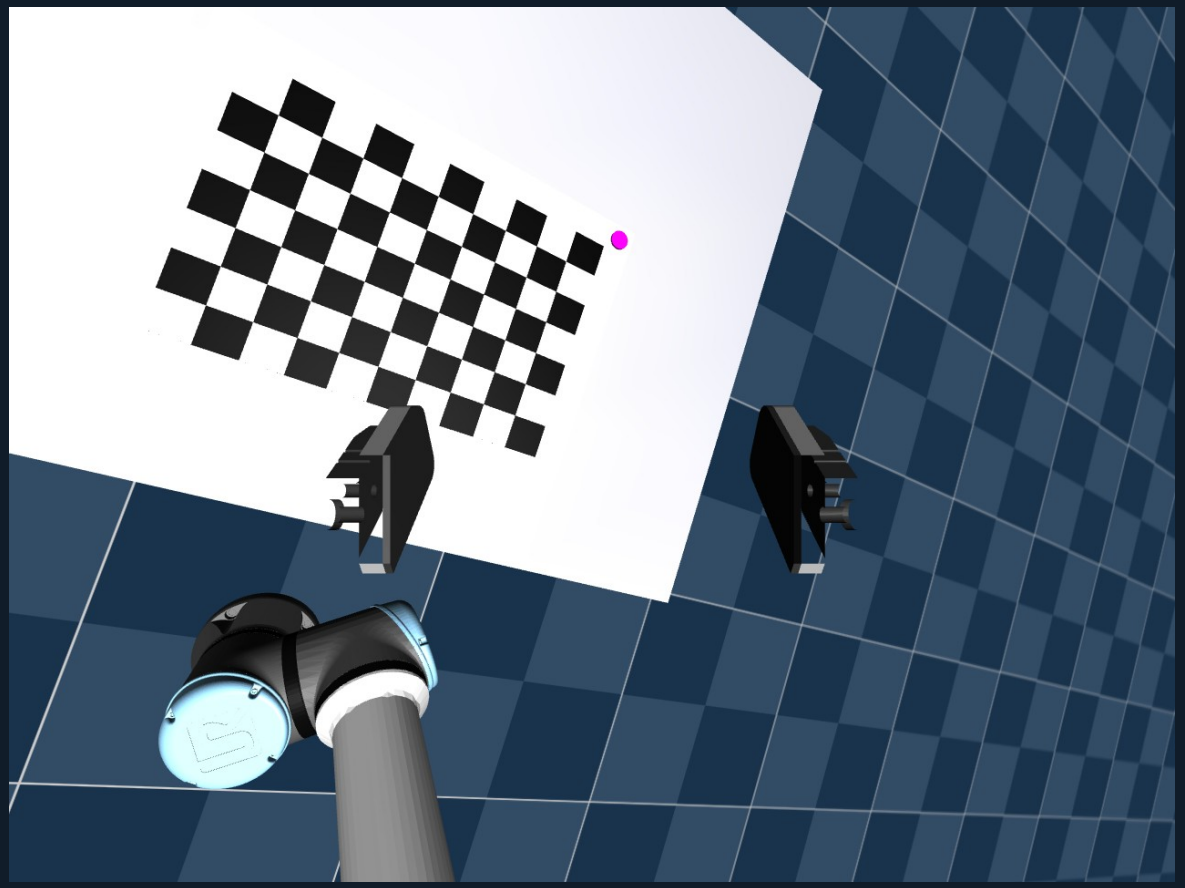


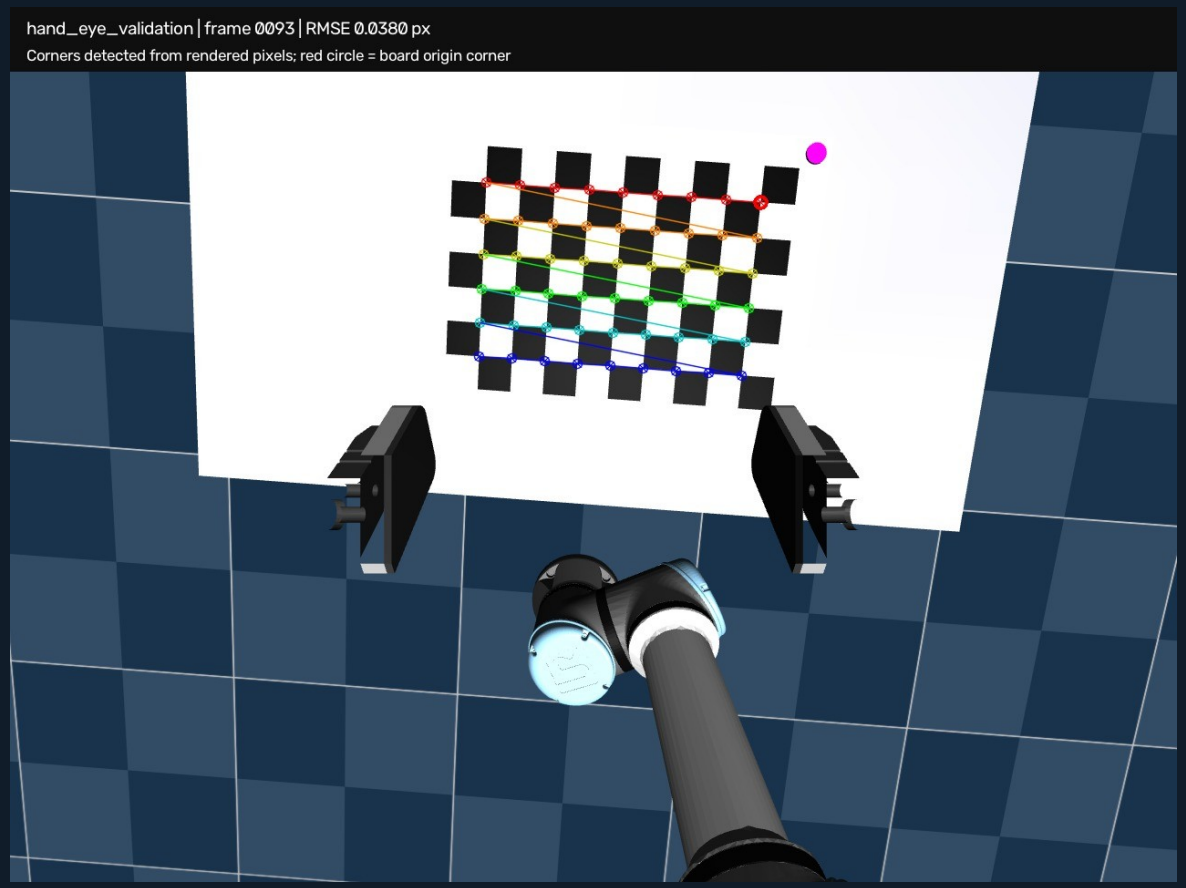


24 training views + 8 held-out views | 0.02177 px validation RMSE

Actual calibration captures | Zhang 2000 | docs/results/wrist-calibration-v1

Estimate the camera–gripper transform



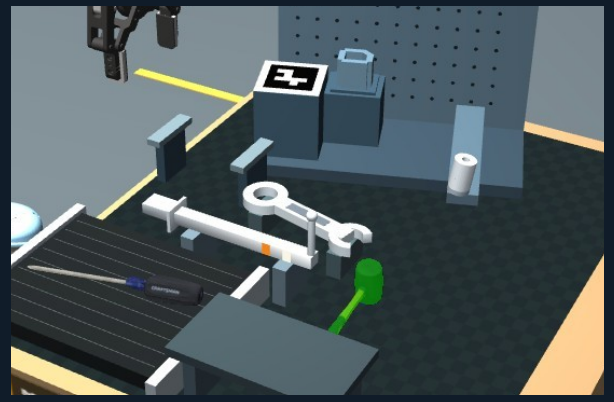


AX = XB | Estimated hand–eye error: 0.852 mm / 0.0227°

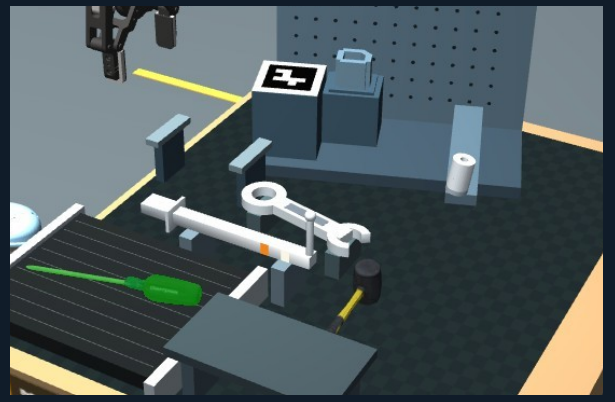
Actual calibration captures | OpenCV AX=XB conventions | 24 training / 10 held-out views

TASK 4 / LOCALIZATION

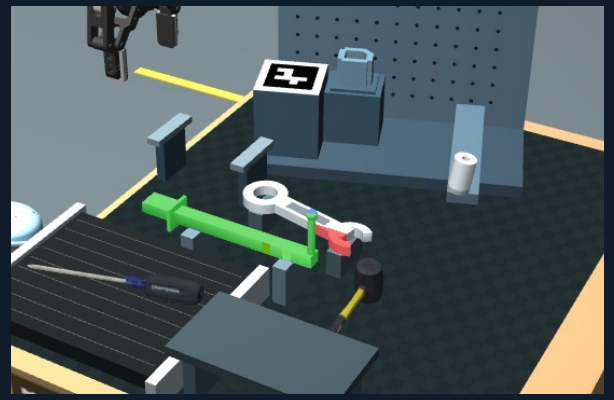
What the foreground detector actually sees



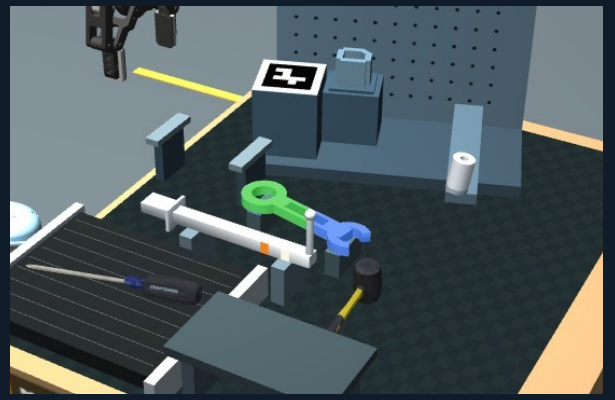
Hammer



Screwdriver



Rack Frame

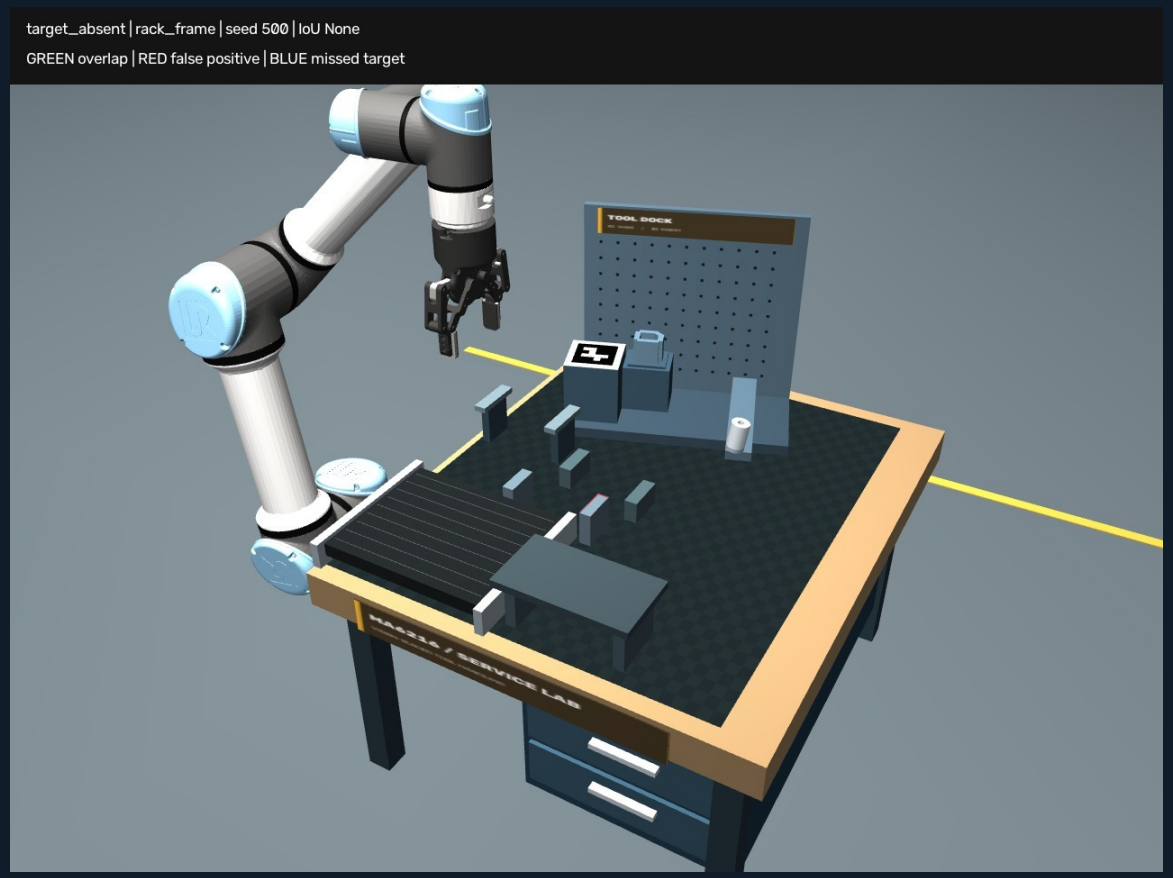


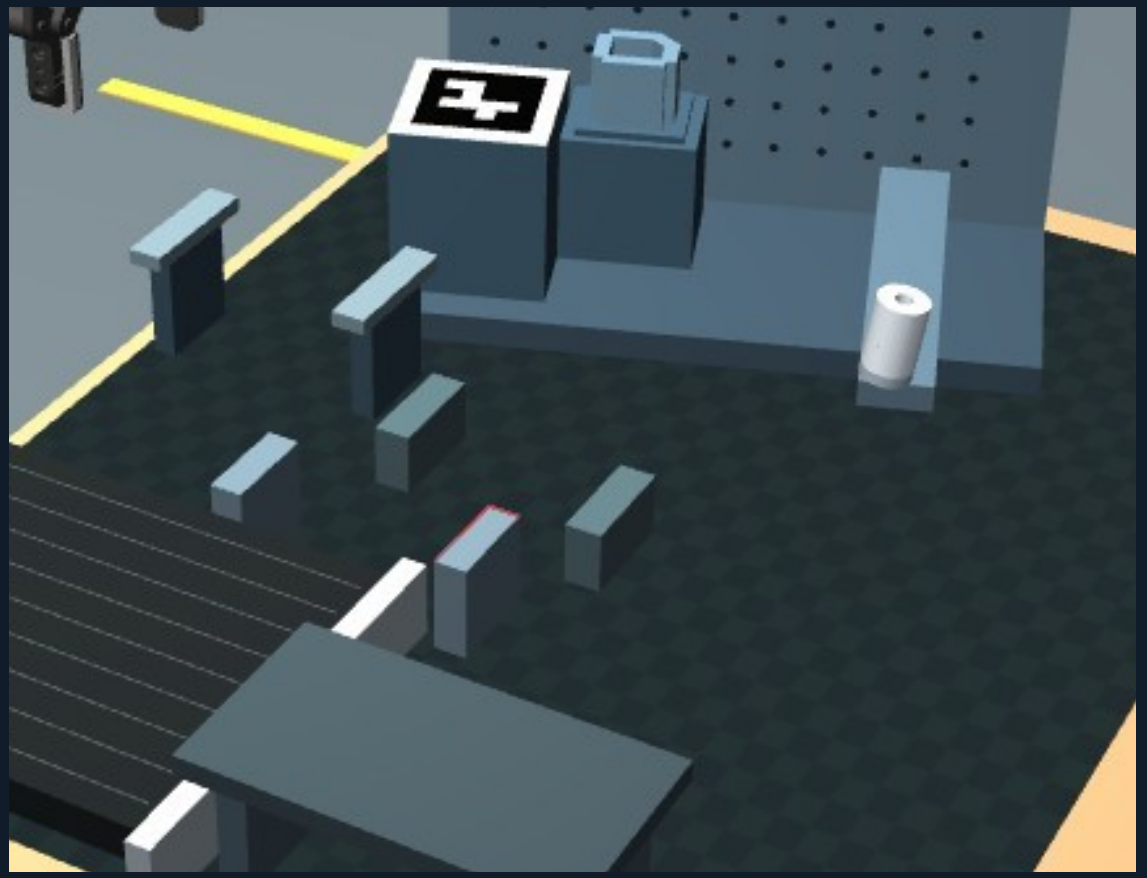
Spanner

50 scenes / 200 queries; predictions saved before ground-truth scoring.

Localization audit | nominal seed 500 | green: overlap; red: extra; blue: missed | cropped original overlays

Failure: background geometry becomes foreground



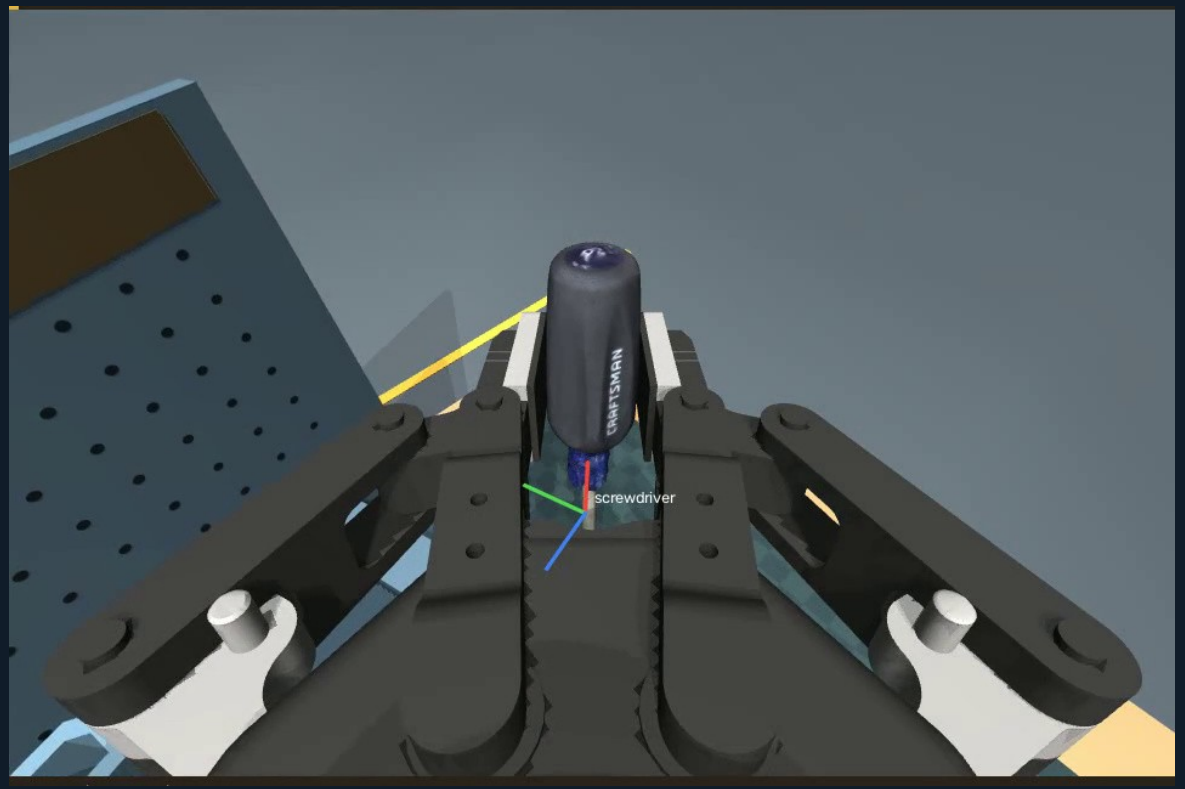


The red region is a false positive, even though no rack target is present.

Separate development diagnosis | target absent / rack frame / seed 500 | not part of final 48 manipulation trials

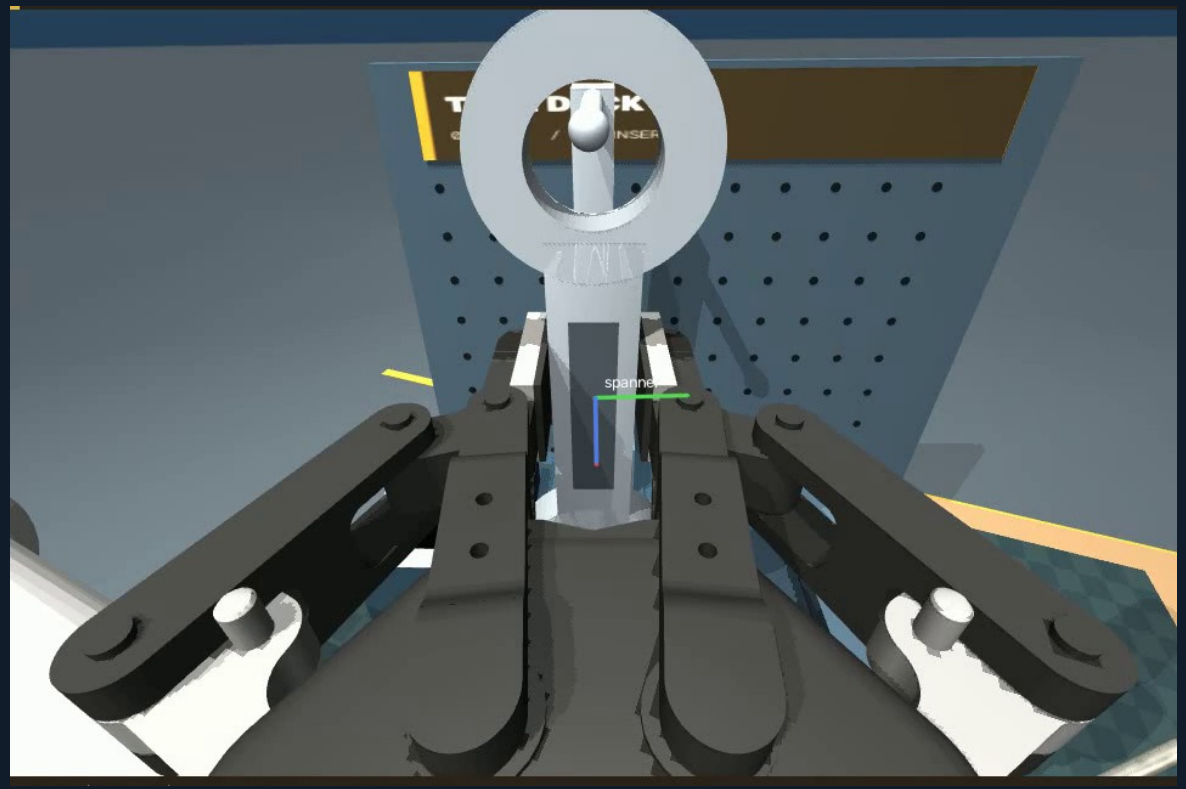
6D pose must preserve the shaft and the hole axes

SHAFT → tilted socket



WRIST / VIDEO 54.40 s

HOLE AXIS → hanging pin

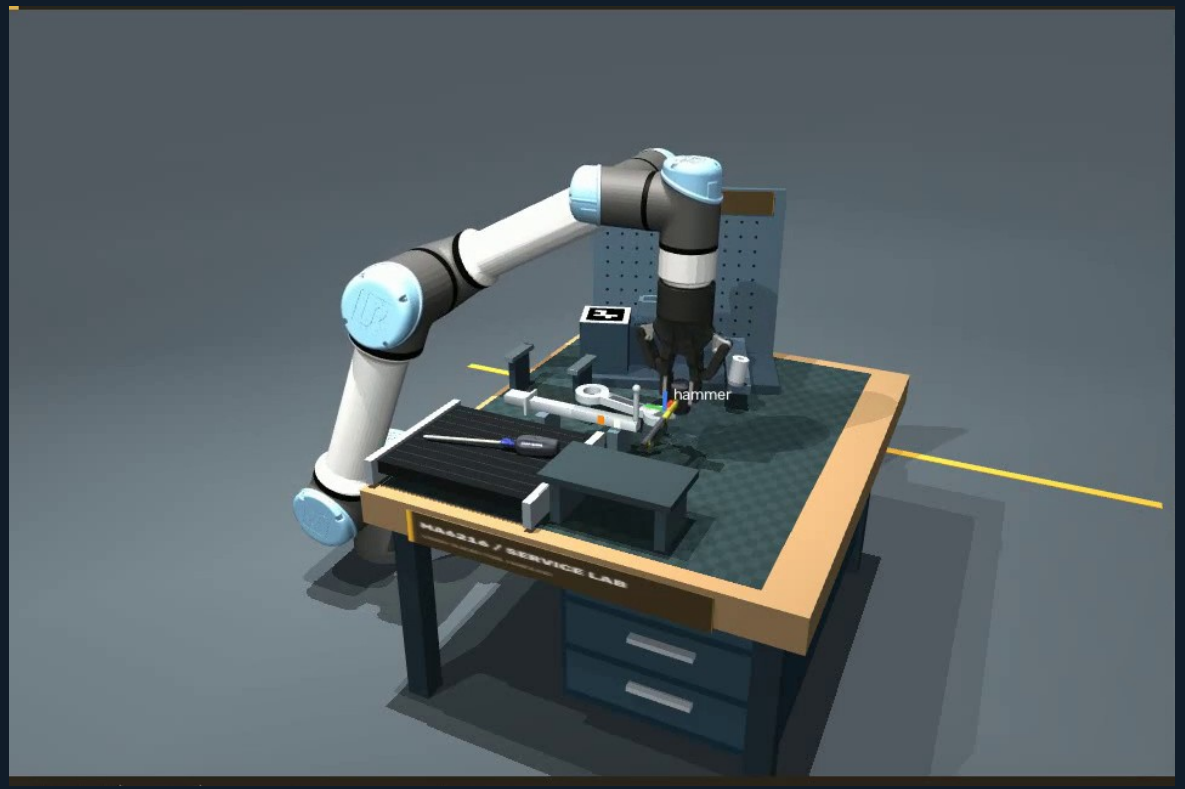


WRIST / VIDEO 171.00 s

The wrist estimates connect tool geometry to the required final pose.

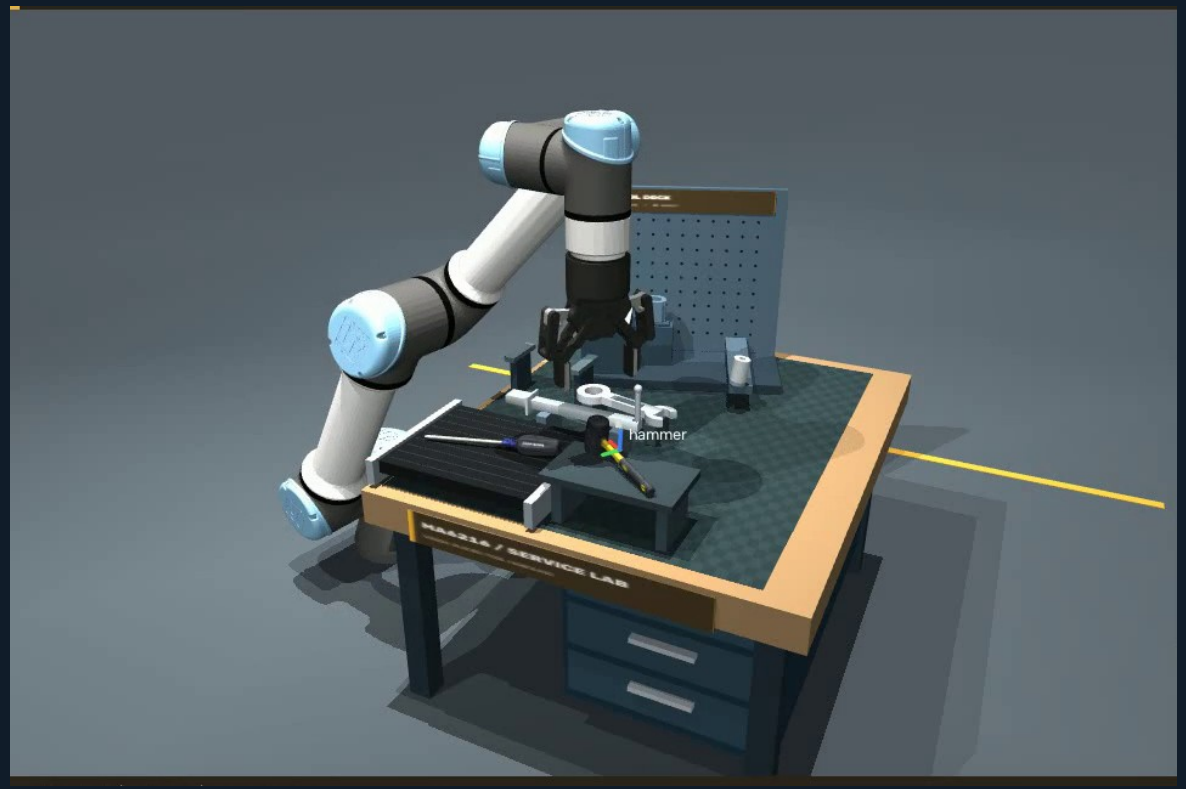
Return the hammer before the next tool arrives

13.60 s / lift from the workbench



OVERVIEW / VIDEO 13.60 s

25.00 s / release into the tray

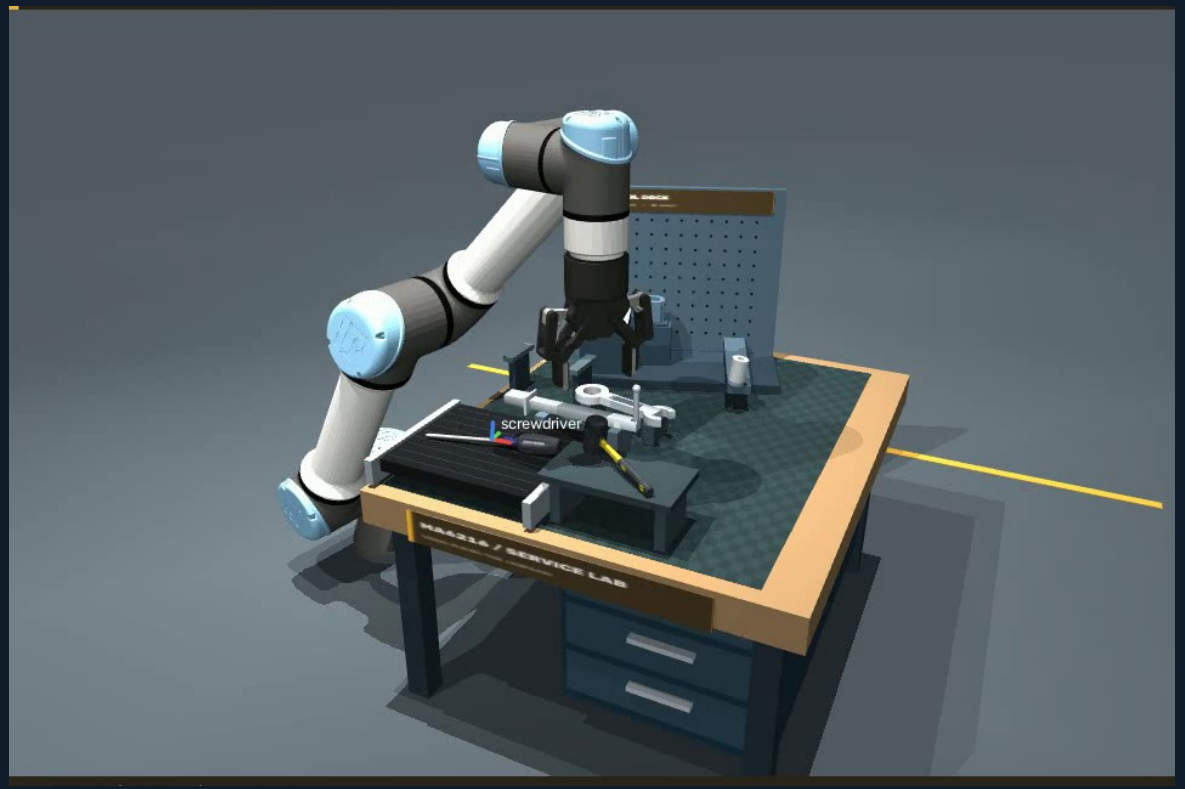


OVERVIEW / VIDEO 25.00 s

Grasp → transfer → release → check stable support.

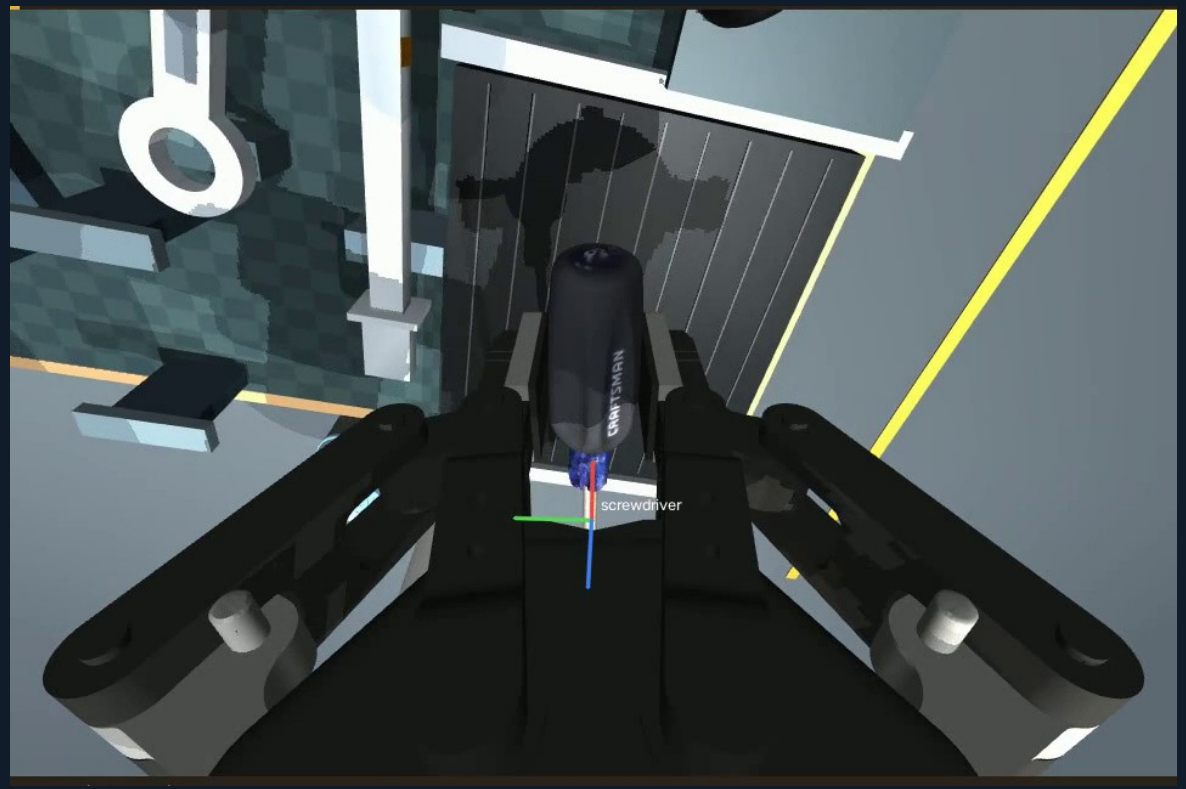
Intercept a tool that is already moving

29.00 s / observe the moving feed



OVERVIEW / VIDEO 29.00 s

43.40 s / tool lifted in the gripper

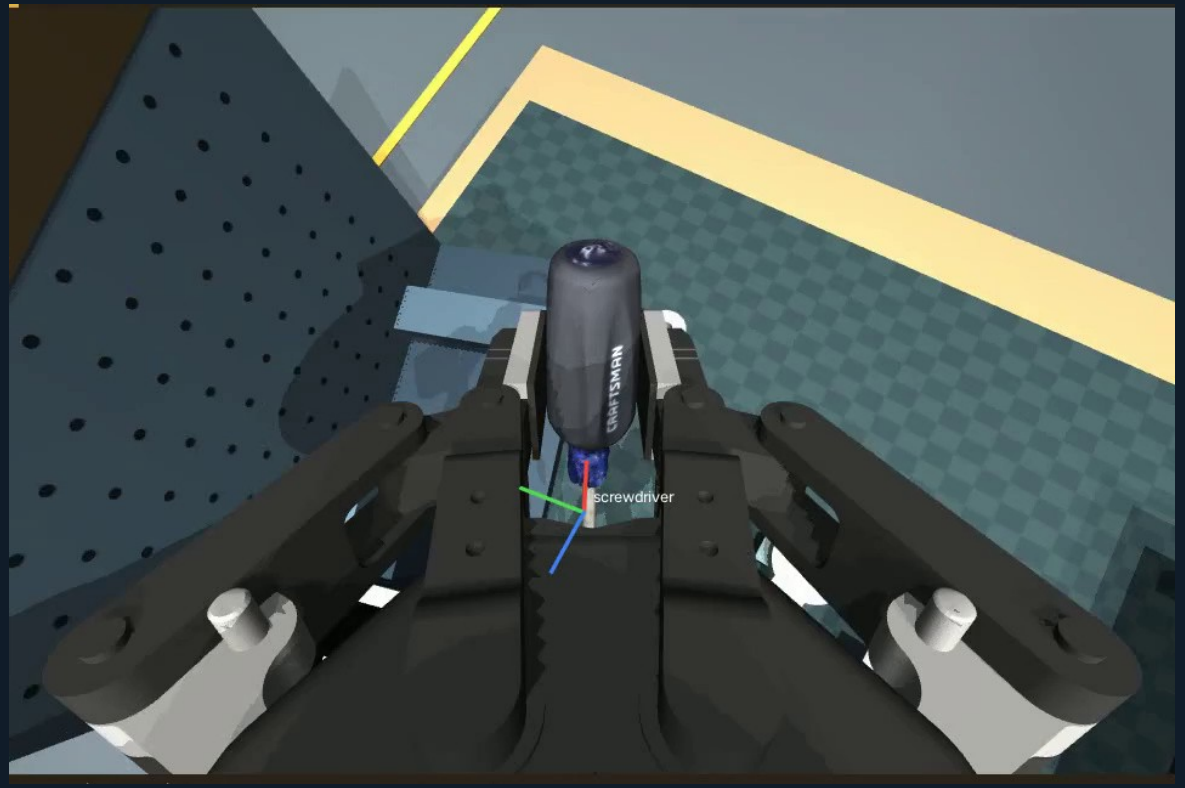


WRIST / VIDEO 43.40 s

Prediction and pickup contact both matter before alignment begins.

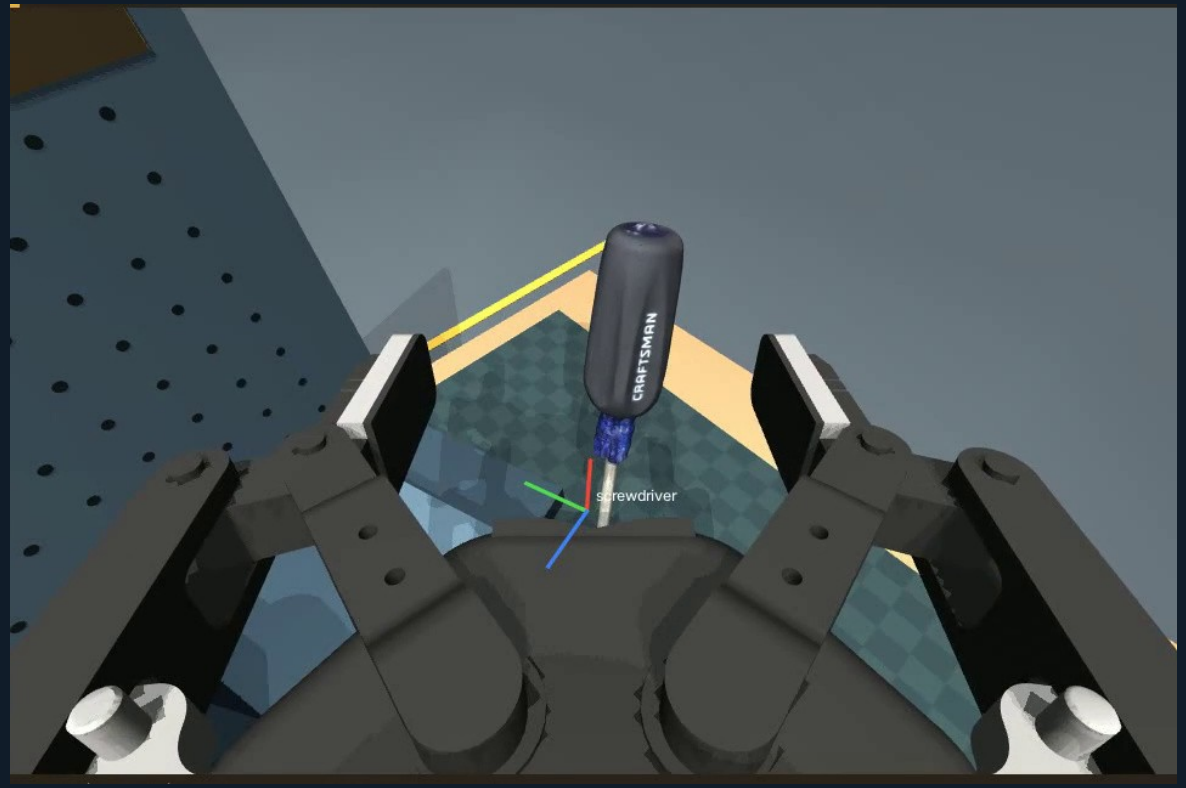
Rotate the shaft, then insert into a tilted socket

50.00 s / reorient the held screwdriver



WRIST / VIDEO 50.00 s

67.00 s / release after insertion

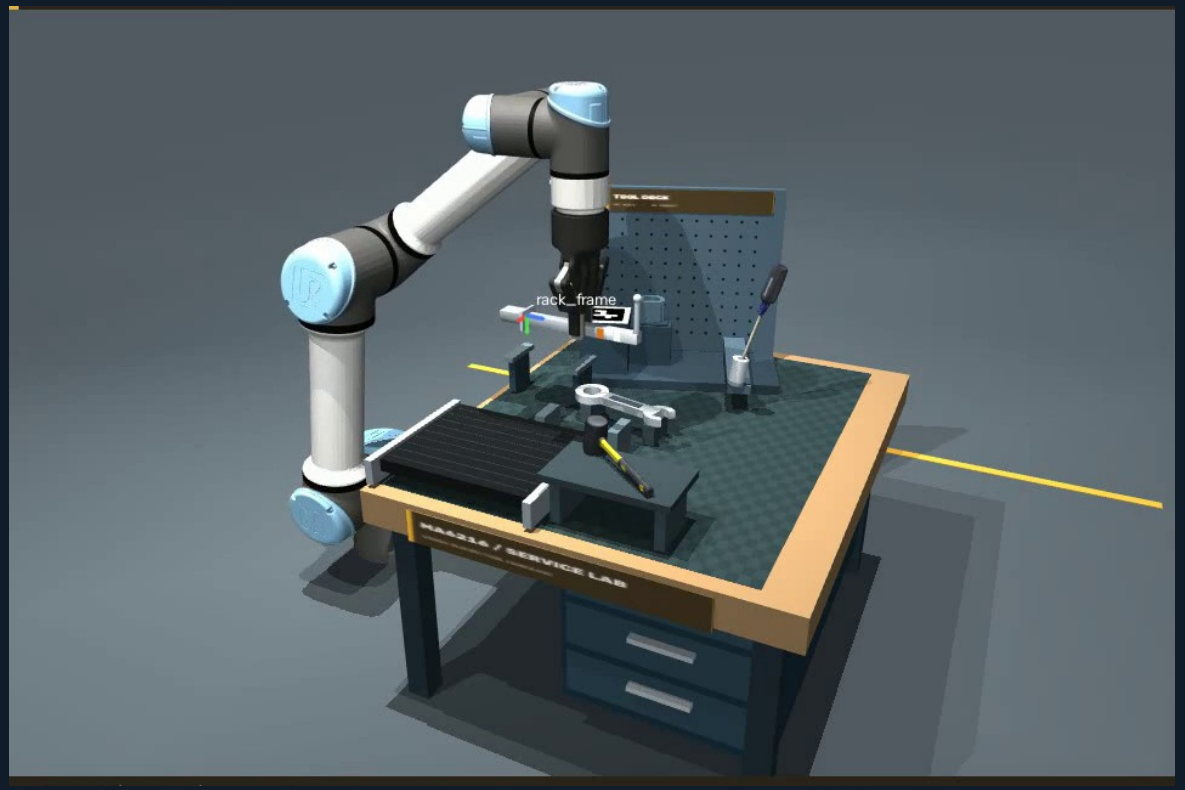


WRIST / VIDEO 67.00 s

Align the tool axis before insertion.

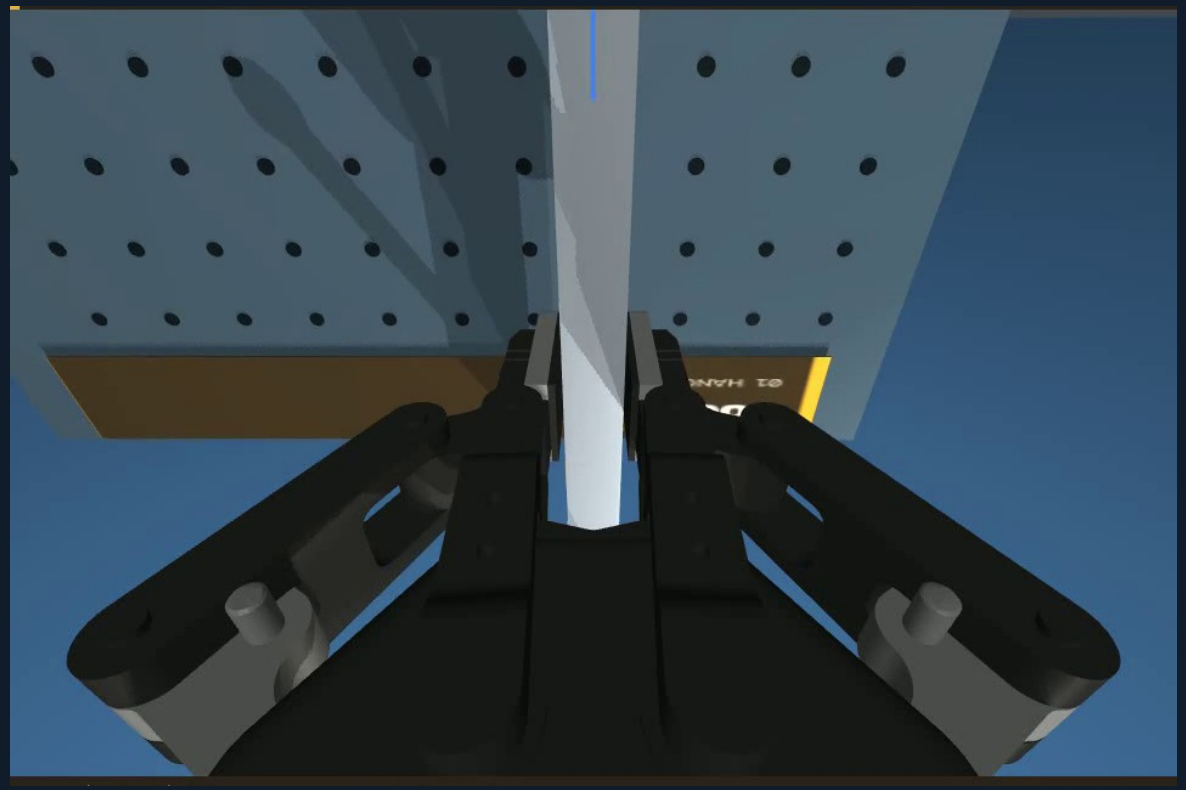
Install the rack as a separate physical object

91.00 s / lift the independent frame



OVERVIEW / VIDEO 91.00 s

112.00 s / seat the frame on its base



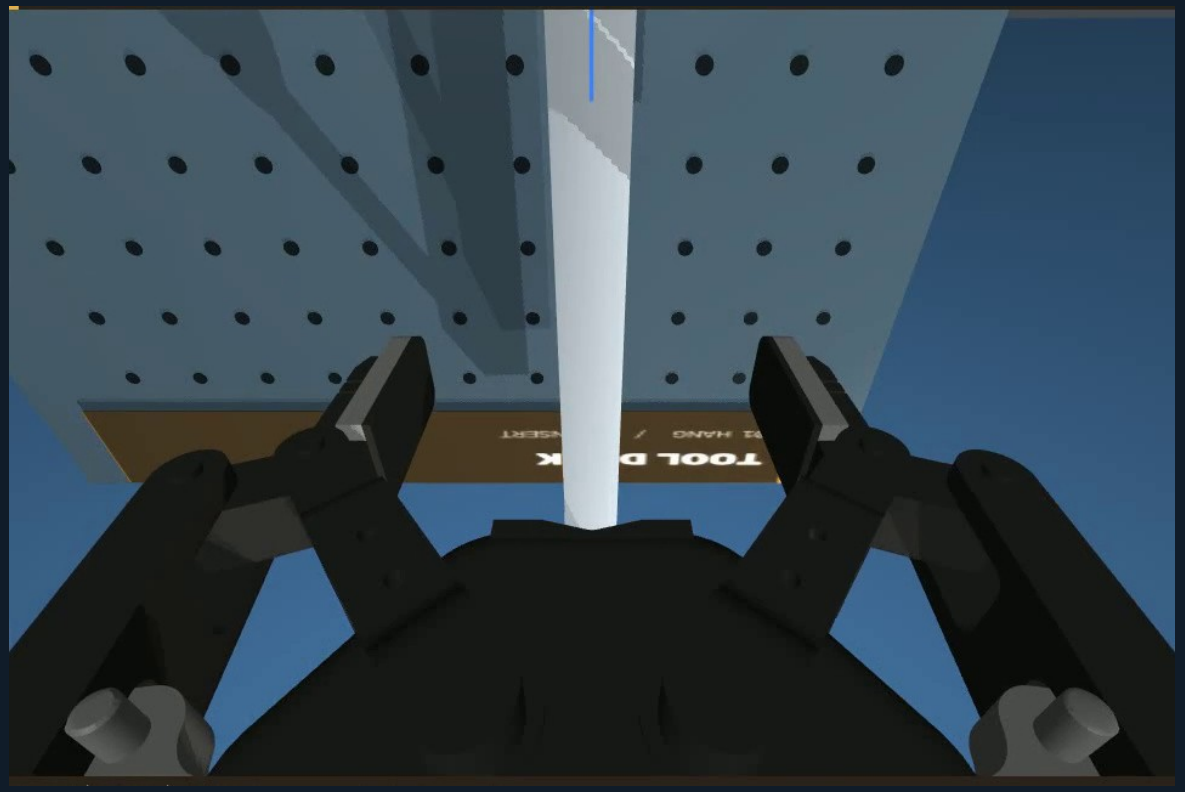
WRIST / VIDEO 112.00 s

Install and release the rack before using it as a fixture.

OPERATION 03 / RELOCALIZATION

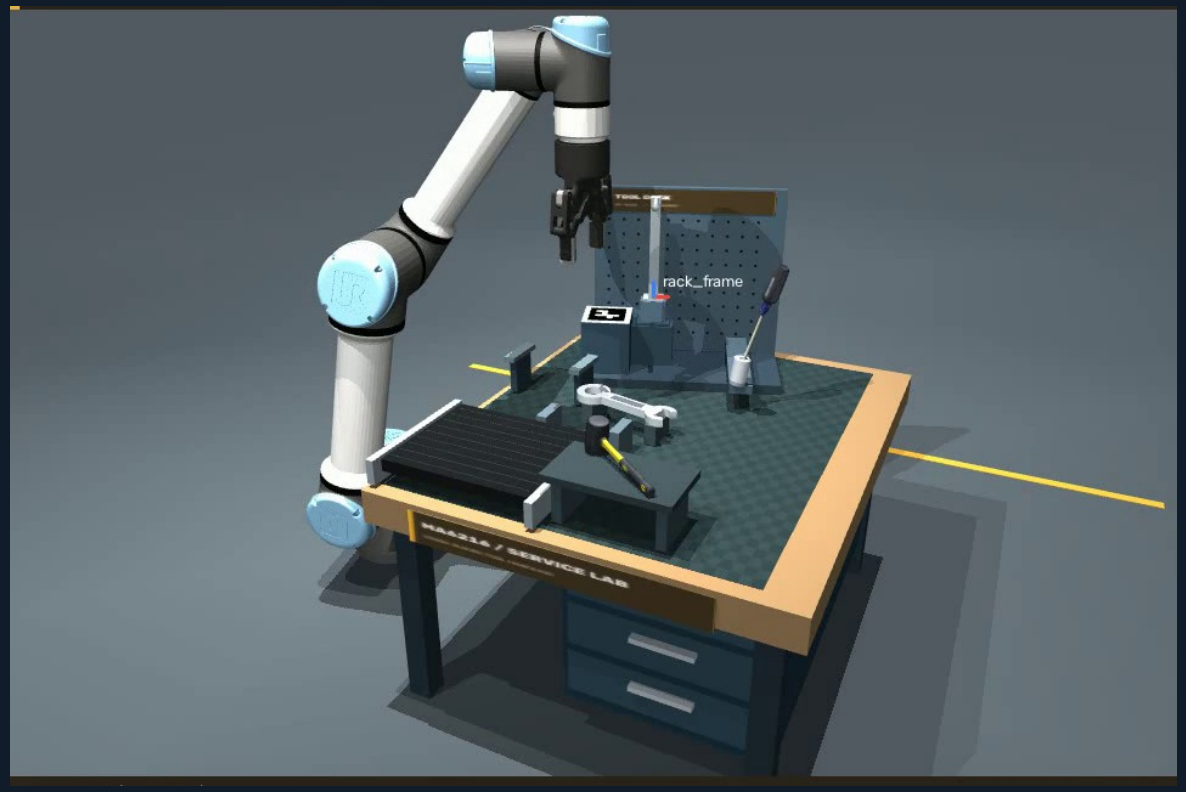
Release the rack, then observe the assembled scene

118.00 s / gripper withdrawal



WRIST / VIDEO 118.00 s

134.40 s / assembled-rack check



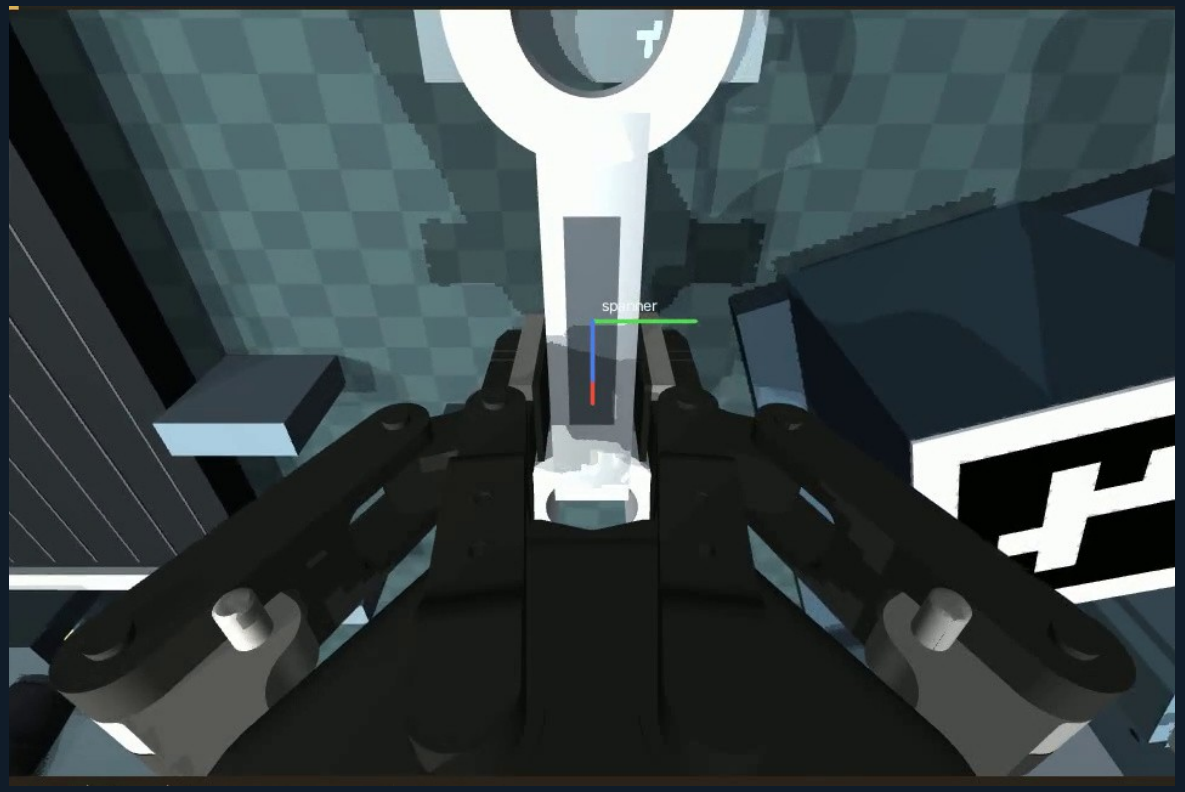
OVERVIEW / VIDEO 134.40 s

Reobserve the rack after physical seating.

Final 48 trials | iterative / no disturbance / seed 400 | playback timestamps

Put the spanner down and actually open the fingers

147.00 s / supported placement



WRIST / VIDEO 147.00 s

150.40 s / gripper clear of the tool

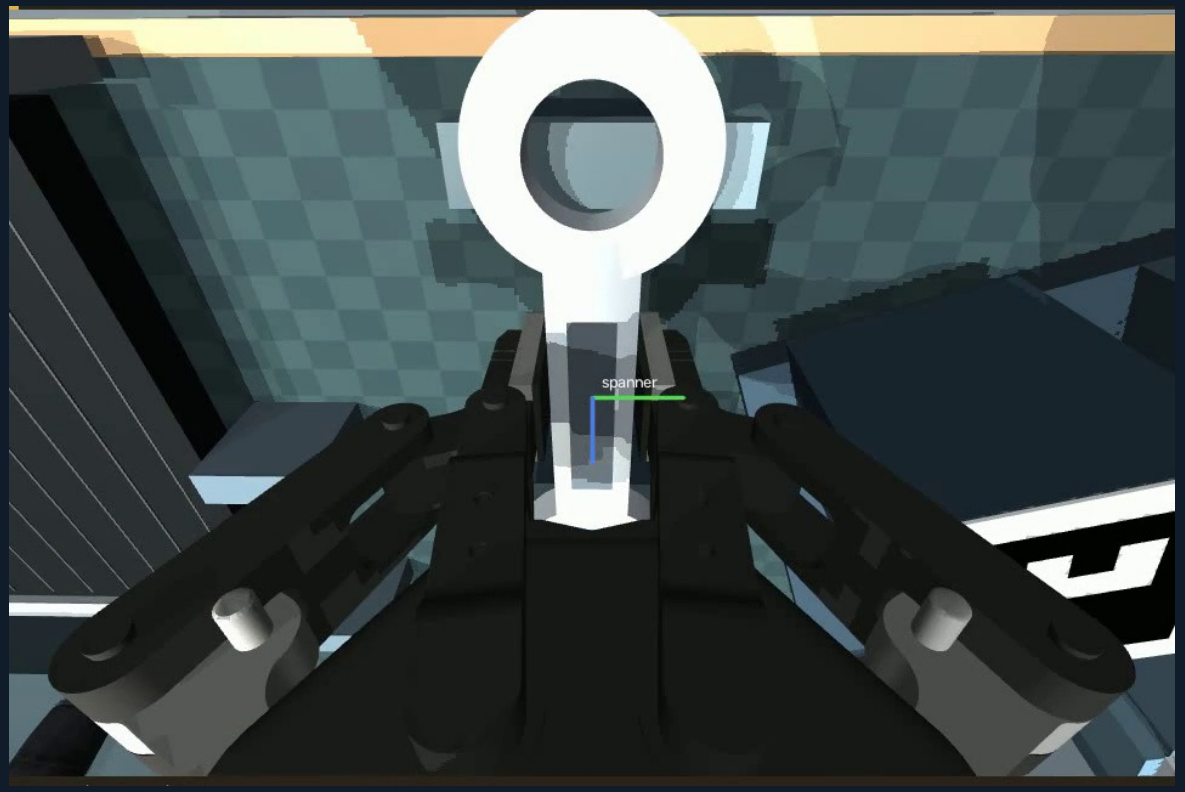


WRIST / VIDEO 150.40 s

The change of grasp includes an explicit release and a new observation.

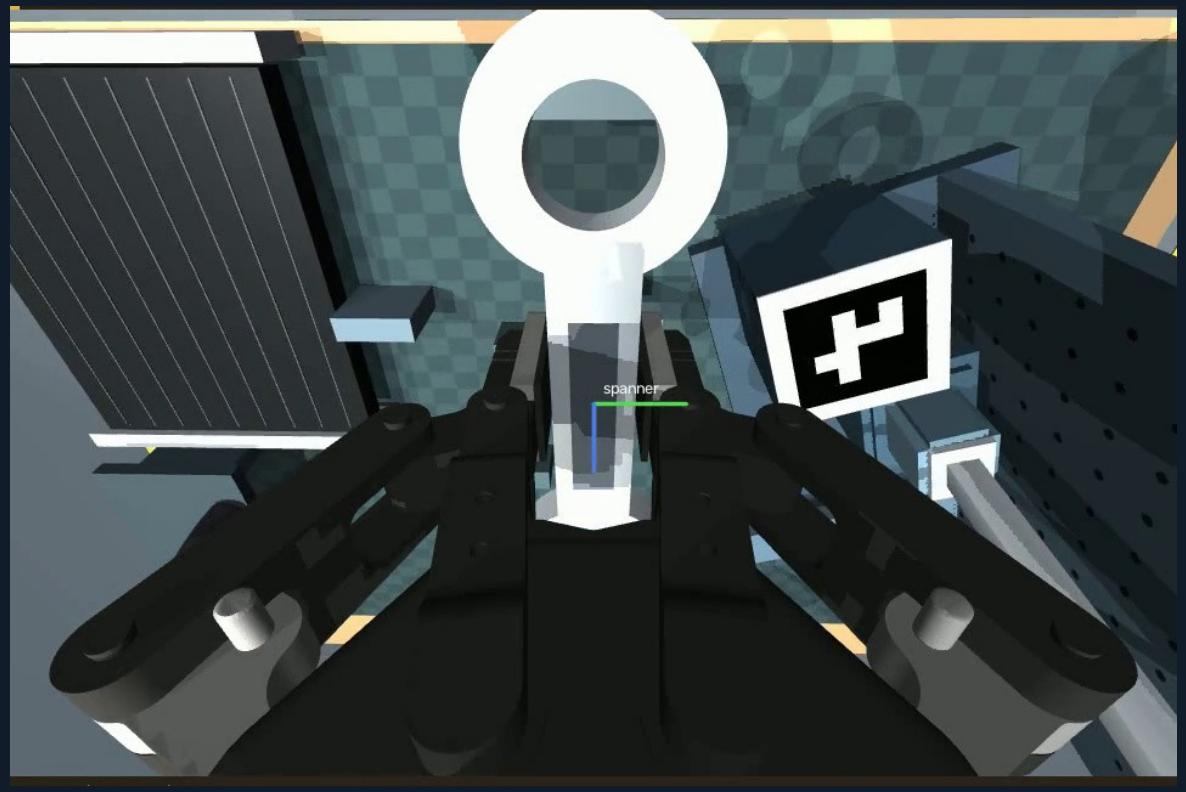
Regrasp so the hole can reach the hanging pin

155.60 s / close at the new grasp



WRIST / VIDEO 155.60 s

158.20 s / lift and inspect

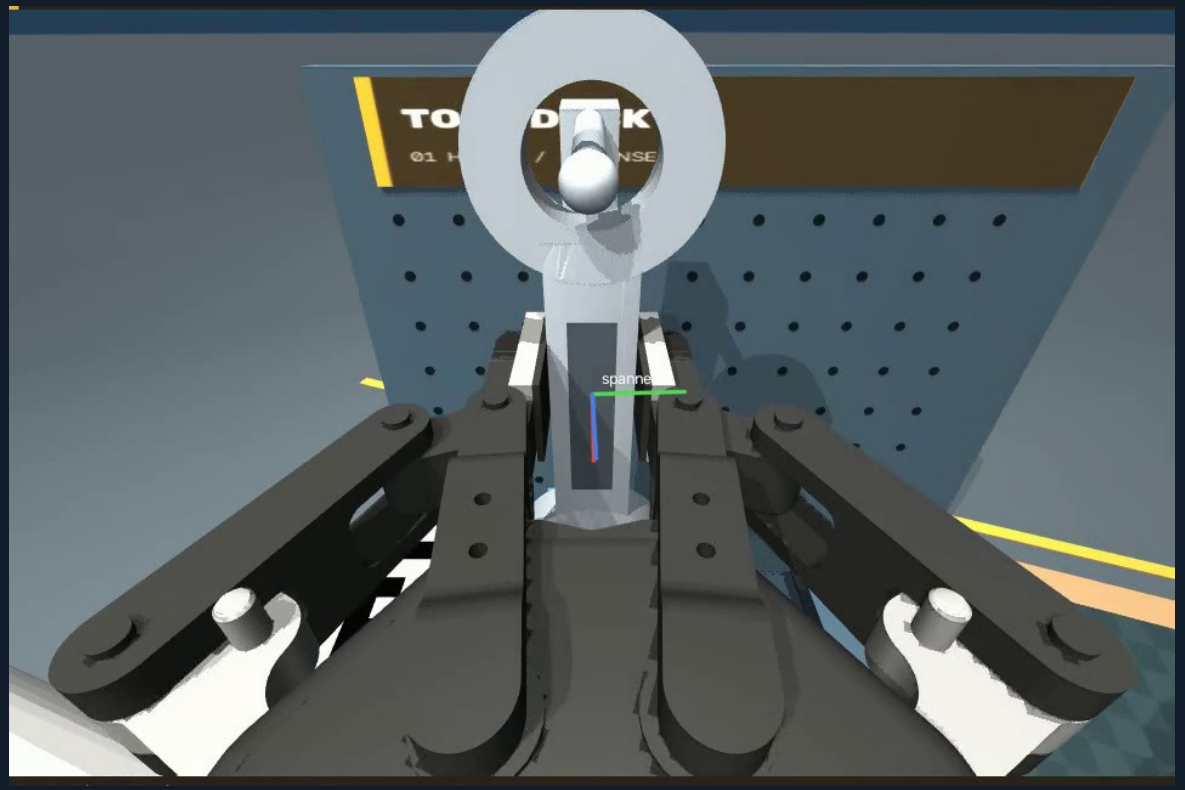


WRIST / VIDEO 158.20 s

The second grasp makes the hole accessible for the final approach.

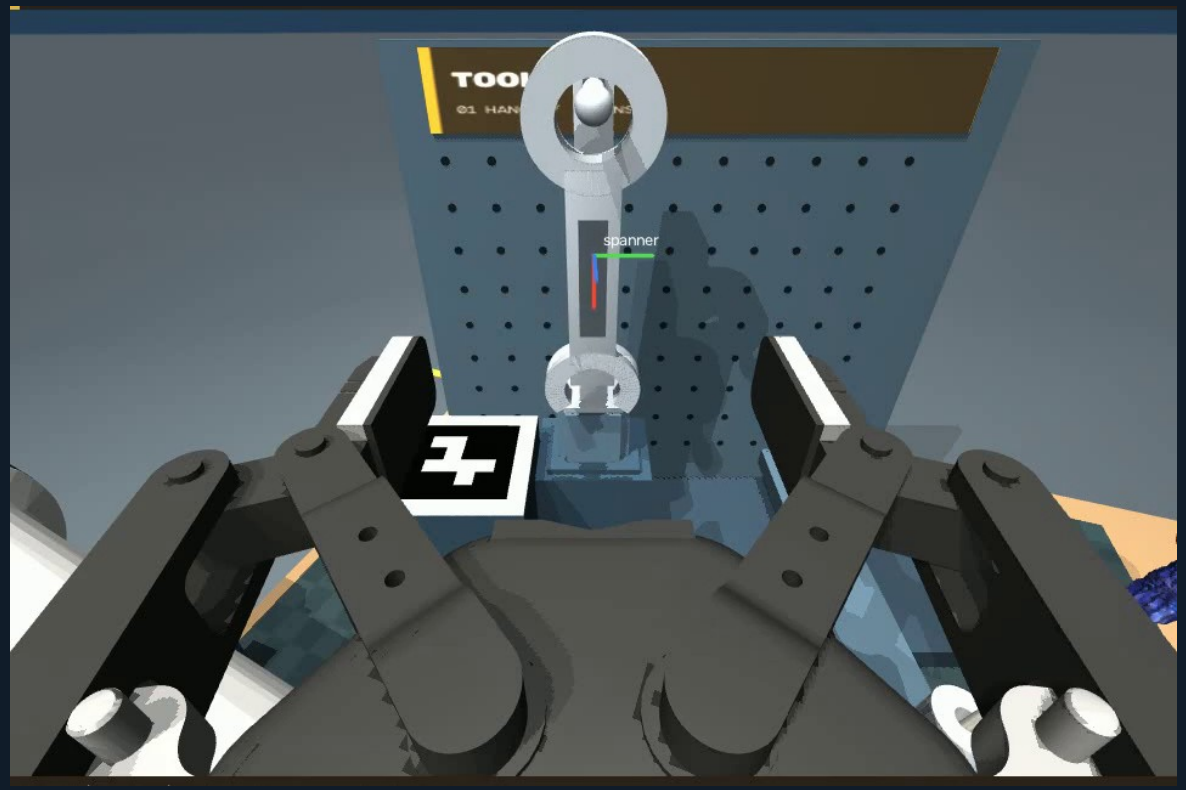
Thread the hole, release, and leave the tool hanging

179.00 s / thread the eye onto the pin



WRIST / VIDEO 179.00 s

195.00 s / released final state



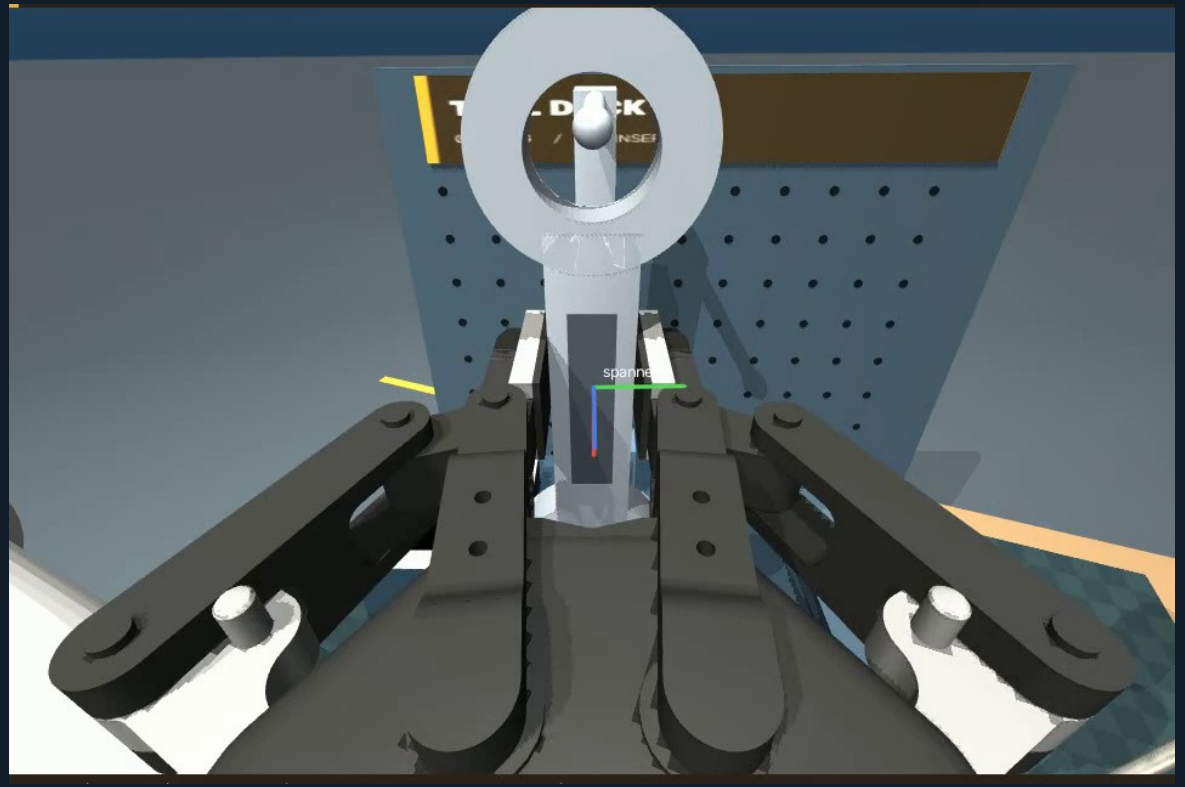
WRIST / VIDEO 195.00 s

Success requires support by the rack after the gripper withdraws.

BONUS / CORRECTION EXAMPLE

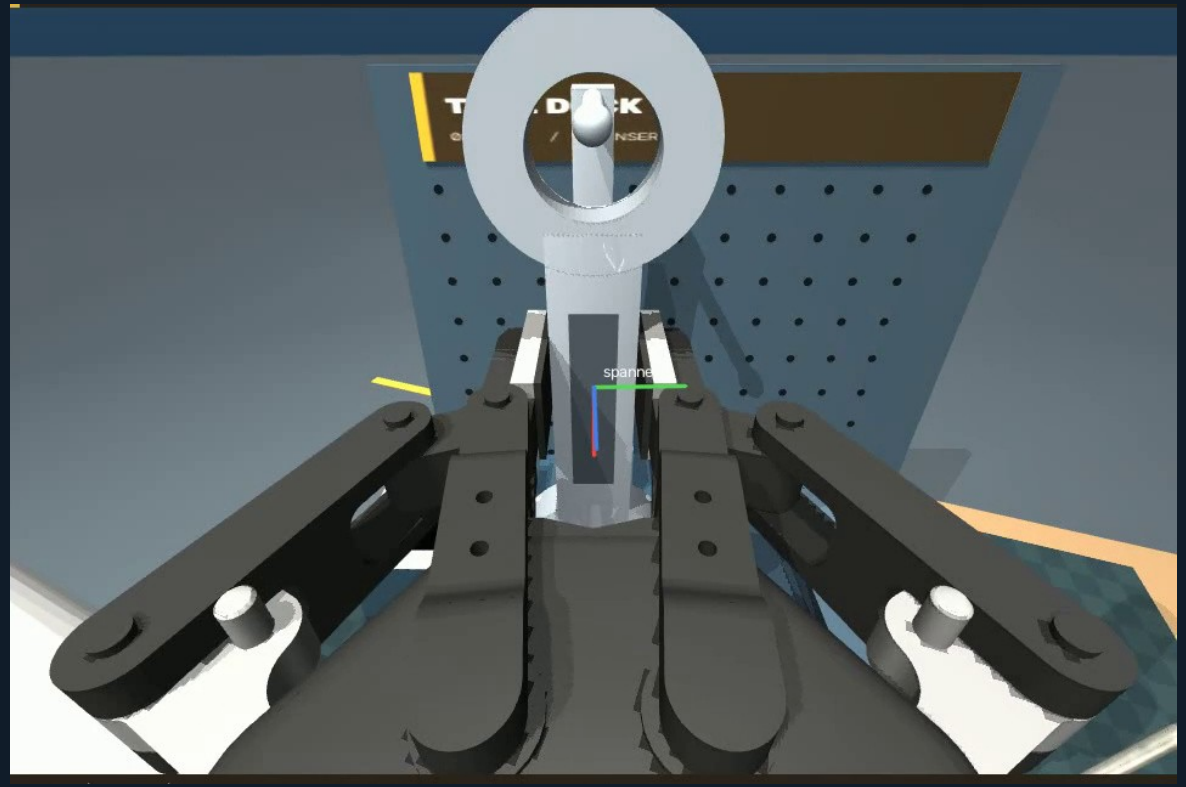
Visual feedback can trigger a real corrective motion

BEFORE / visual angular residual 1.60°



WRIST / VIDEO 171.64 s

AFTER / visual angular residual 0.74°



WRIST / VIDEO 172.76 s

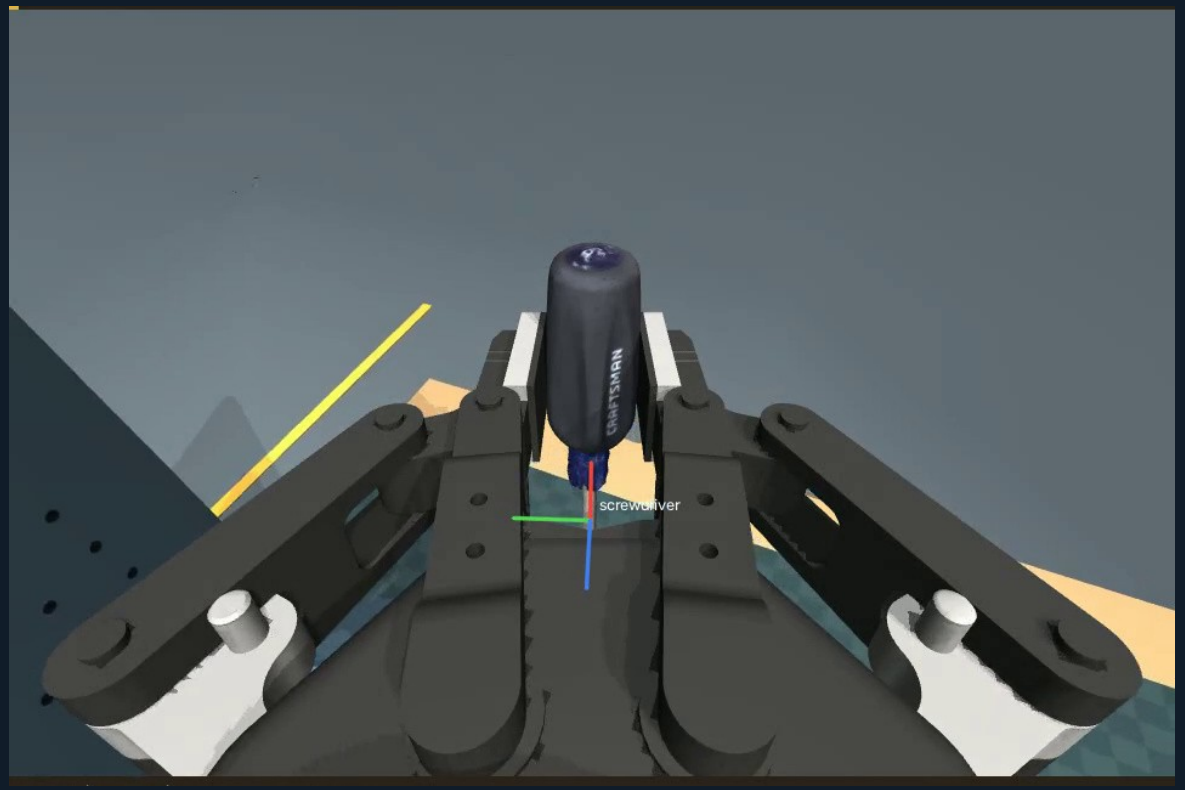
One logged correction; a separate integration example.

Separate calibration integration / iterative / alignment impulse / seed 400 | original log residuals

PAIRED EVIDENCE / SEED 400

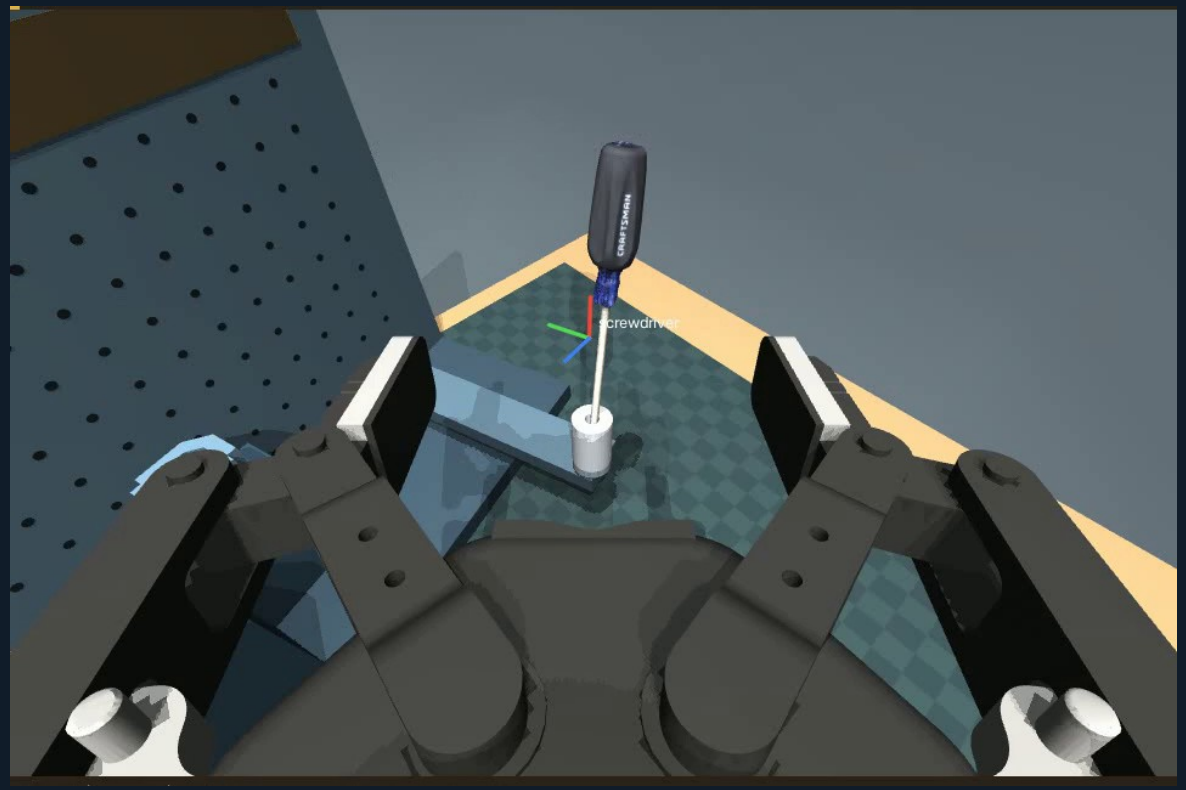
Same initial seed, different insertion outcomes

OPEN LOOP / insertion guard stops the run



WRIST / VIDEO 57.96 s

ITERATIVE / driver released and stable



WRIST / VIDEO 70.00 s

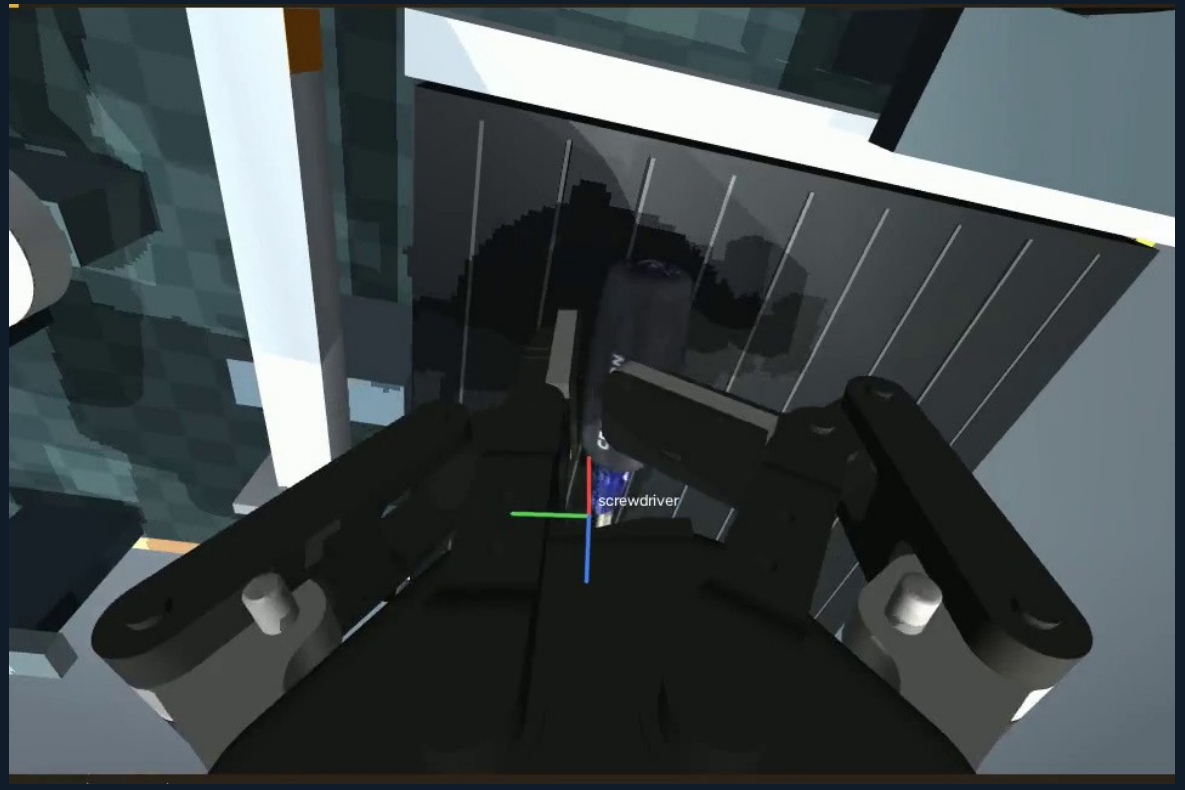
A matched example, not a claim of overall superiority.

Final 48 trials | open_loop vs iterative / no disturbance / seed 400 | different playback times

FAILURE EVIDENCE / SEED 401

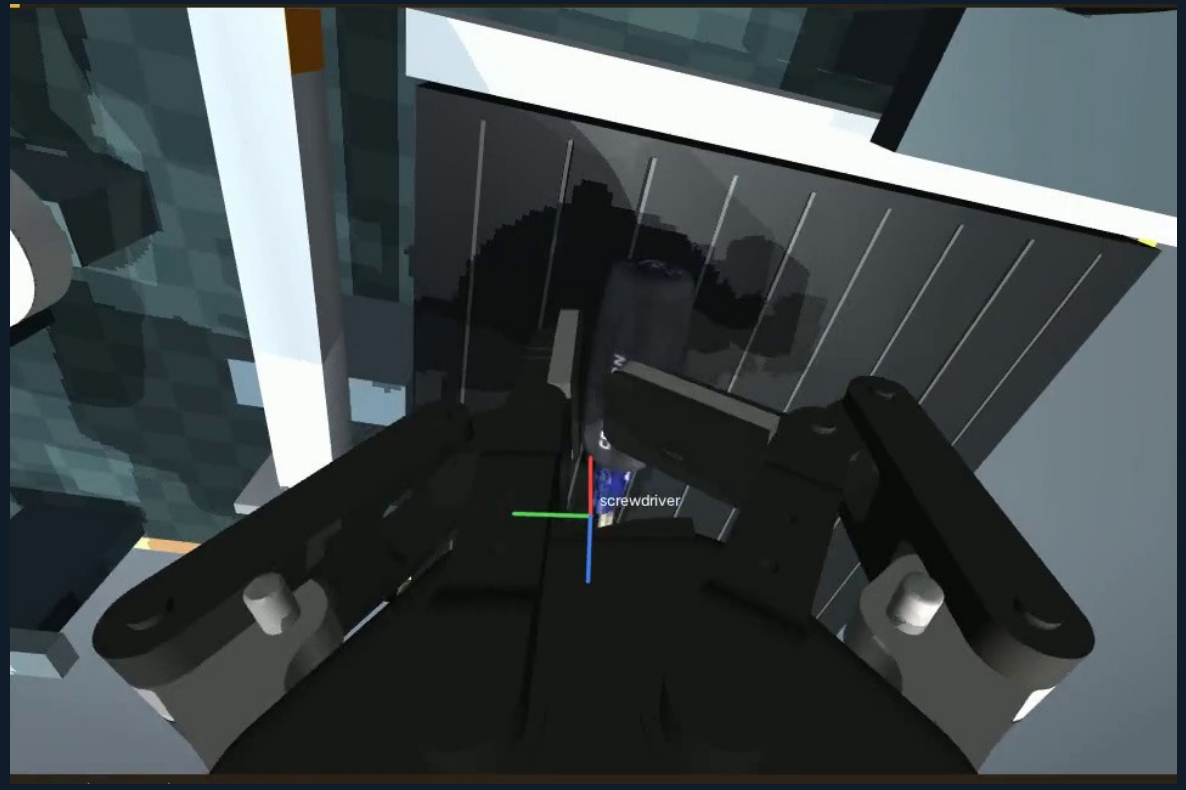
Early pickup failure defeats both feedback modes

STAGED / stops at gripper closure



WRIST / VIDEO 40.24 s

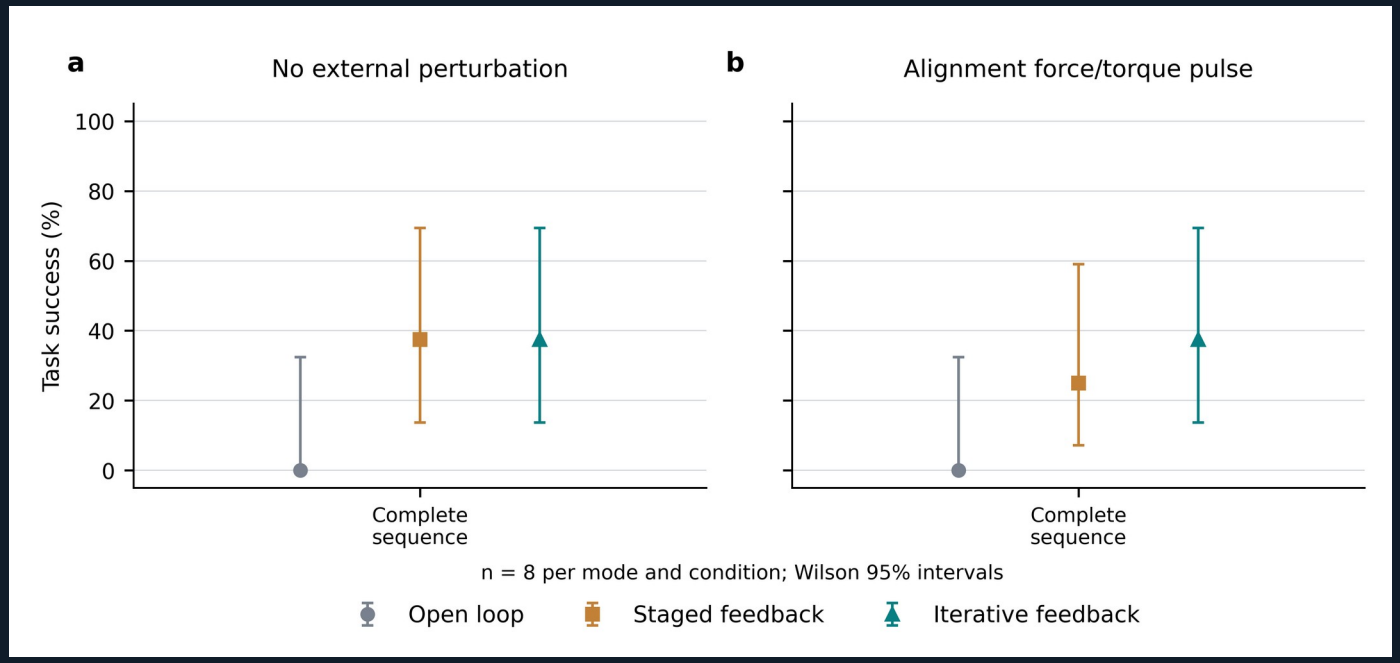
ITERATIVE / also stops at gripper closure

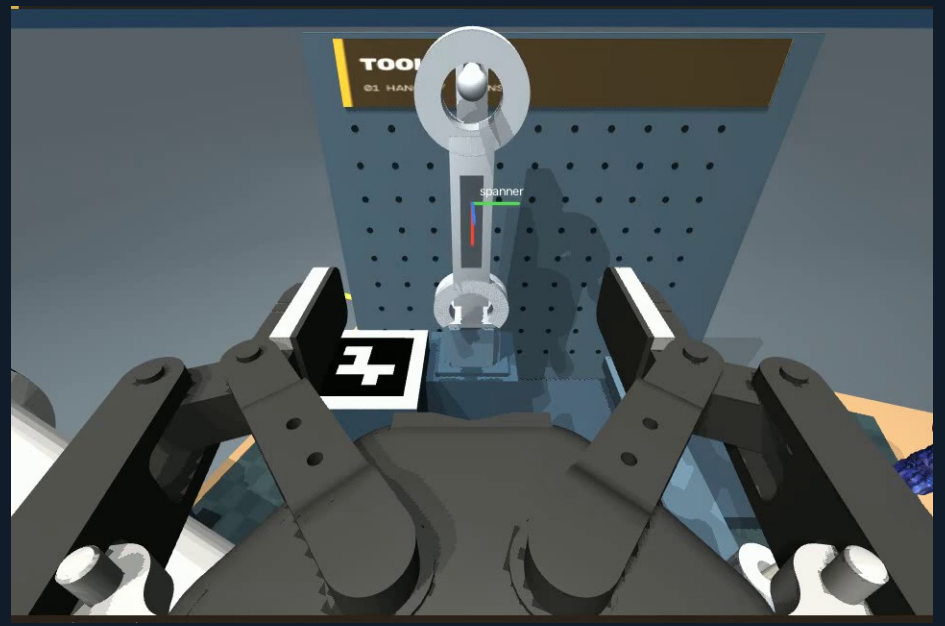


WRIST / VIDEO 40.24 s

Both stop before the later visual alignment loop can help.

All attempts count, including the failed pickups





WRIST / VIDEO 195.00 s

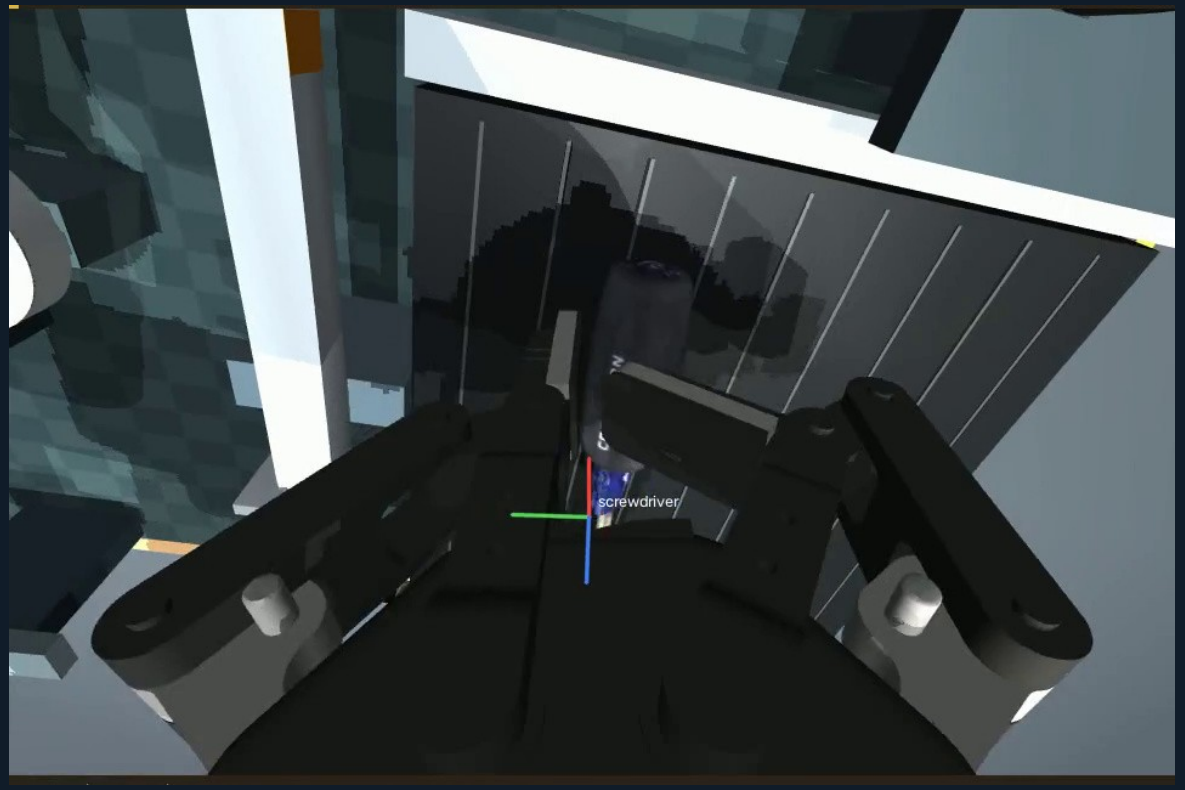
One success image cannot replace all 48 outcomes.

Wide uncertainty; no statistically established advantage in this sample.

ENGINEERING CONCLUSIONS

What the failures tell us to improve next

PICKUP / the dominant bottleneck



WRIST / VIDEO 40.24 s

24 / pickup-contact stops

6 / hammer approach stops

6 / open-loop insertion stops

1 / bilateral-grasp loss

Next: reliable contact → robust visibility

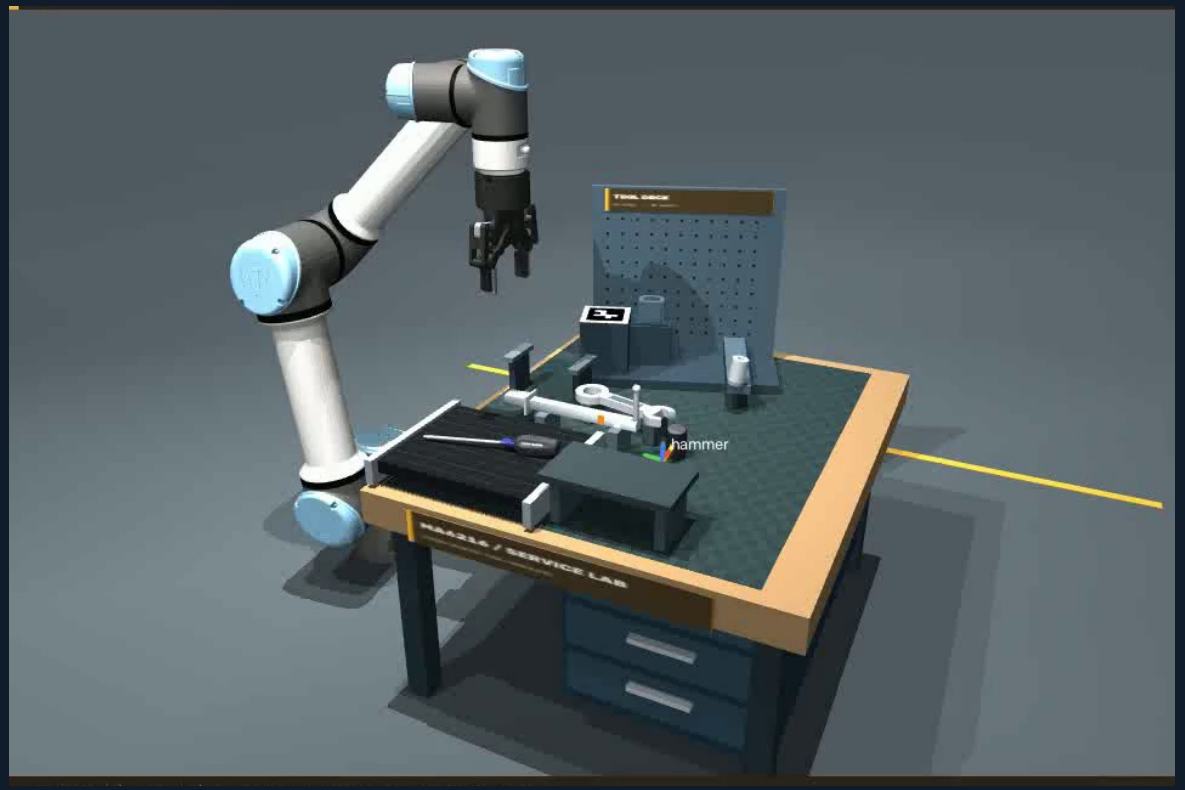
→ broader unseen-scene evaluation

Reliable pickup is required before downstream feedback can help.

CONCLUSION / DISCUSSION

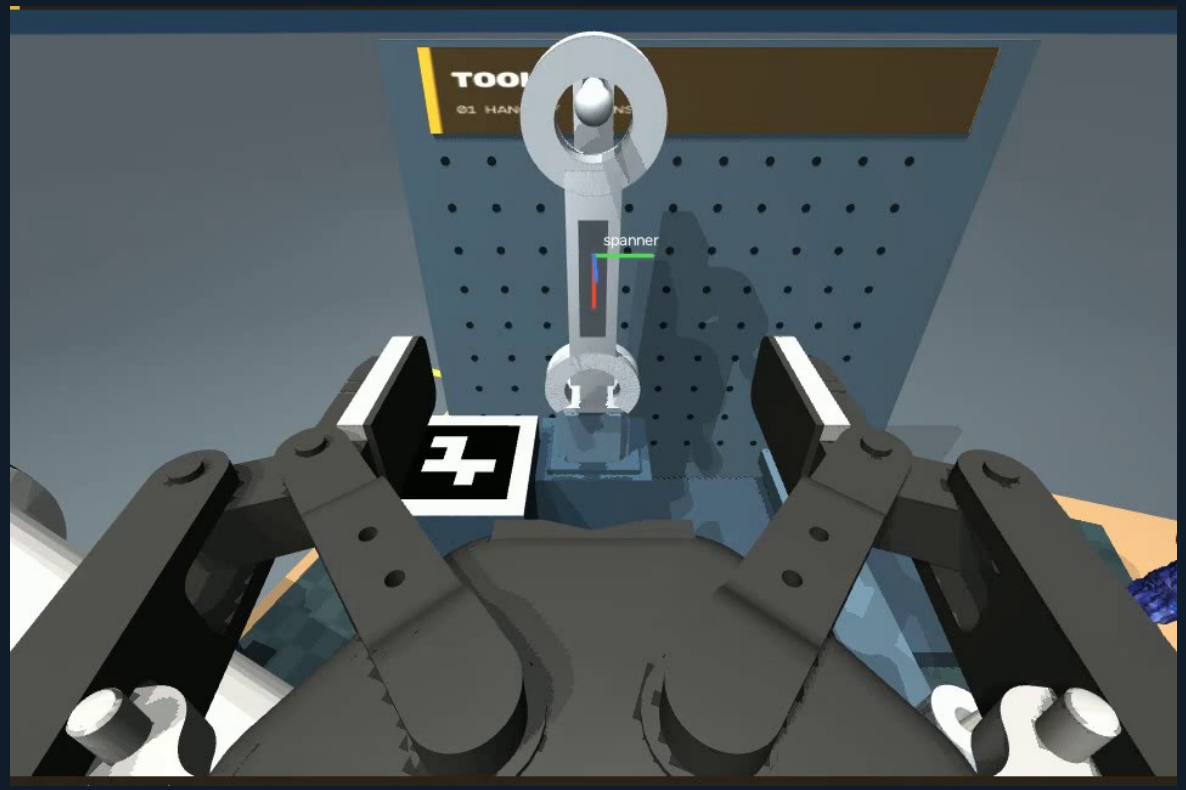
A complete, auditable vision-guided workcell

From calibrated observations



OVERVIEW / VIDEO 0.00 s

To released physical manipulation

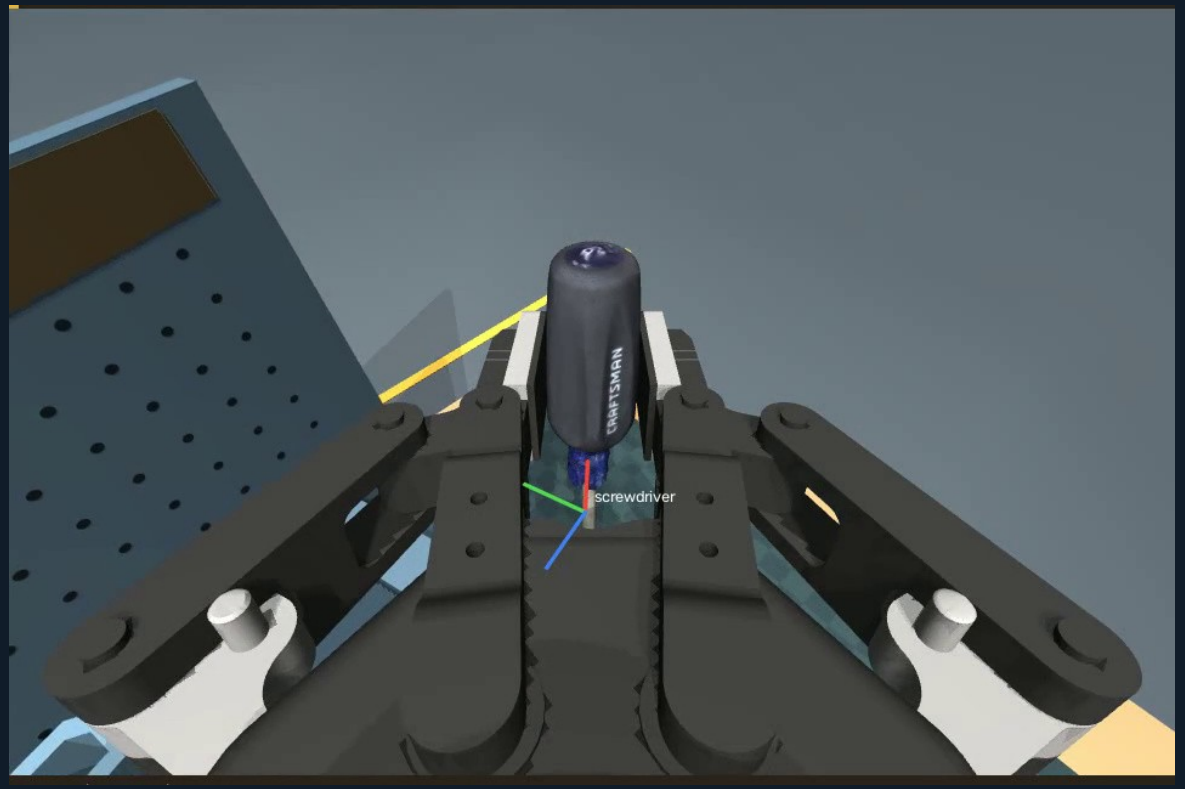


WRIST / VIDEO 195.00 s

Repository evidence: code, calibration, masks, trials and original videos.

Final 48 trials | iterative / no disturbance / seed 400 | playback timestamps

Accuracy, timing and experimental units



WRIST / VIDEO 54.40 s

Initial pose: 2.60 mm / 1.44°

ADD / ADD-S: 3.17 / 2.29 mm

Hand–eye: 0.852 mm / 0.0227°

48 robot attempts ≠ 200 mask queries

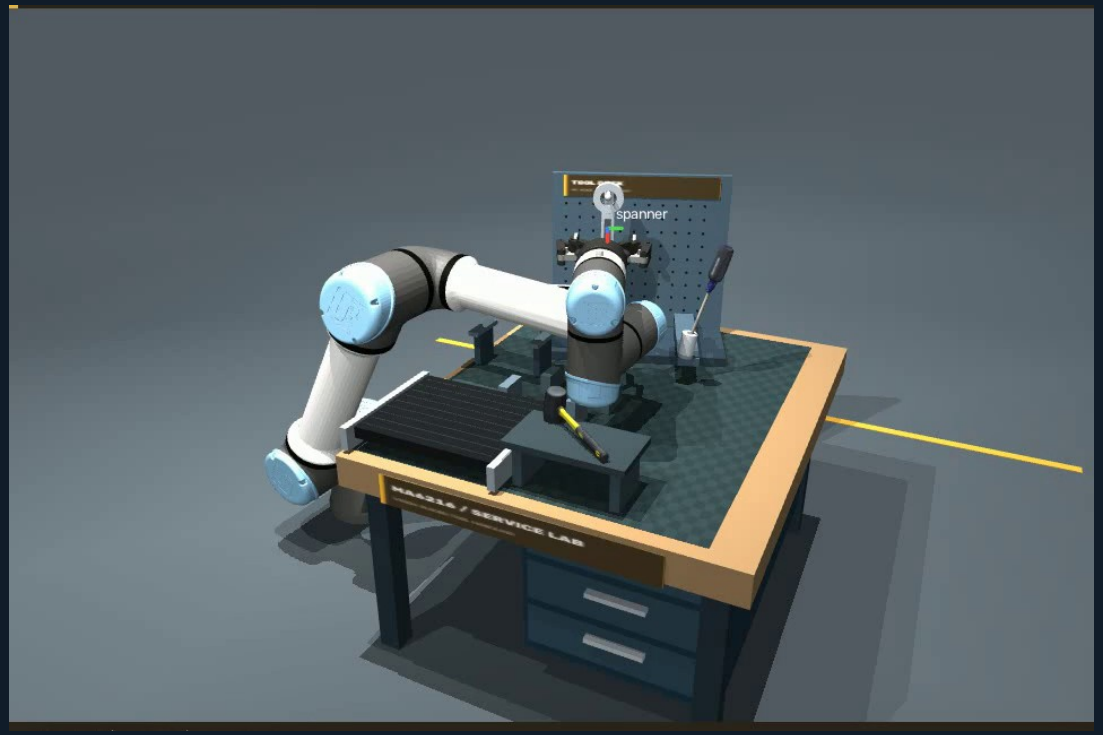
500 Hz physics ≠ visual update rate

Tracking error ≠ final placement error

Do not mix measurement units, denominators or timing definitions.

Full uncertainty and sample counts: docs/results/visual-feedback-calibrated-v1/README.md

Methods and primary references



OVERVIEW / VIDEO 195.00 s

Wen et al. — FoundationPose, CVPR 2024

Zhang — Camera calibration, TPAMI 2000

OpenCV — Hand–eye calibration conventions

Chaumette & Hutchinson — Visual servoing, 2006

Todorov et al. — MuJoCo, IROS 2012

Labbé et al. — MegaPose, 2022

Fang et al. — GraspNet-1Billion, CVPR 2020

Published methods + independent integration + traceable evidence.